Dictionary Learning with Dictionary Size Mismatch

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Abstract

We study the convergence behaviour of the Iterative Thresholding and K-residual Means (ITKrM) algorithm for dictionary learning under a sparse generative signal model with random signs and permutations.

Numerical experiments indicate that dictionary algorithms often converge to stable configurations that do not coincide with the generating dictionary: some atoms are merged, while others appear as duplicates with small deviations. Motivated by these observations and by the fact that in practice the correct number of atoms K is often unknown, we analyse a setting in which the learned dictionary contains one atom fewer than the generating dictionary. Thus we cannot recover all atoms, but analyse the stability of merged configurations we see in experiments. In particular, each stable configuration consists of all generating atoms except that one pair of atoms is covered by a single merged atom of the form $\psi_i \pm \psi_j$ with $i \neq j$. Since the indices i and j are arbitrarily, this gives rise to many distinct stable configurations.

Under standard sparsity and incoherence assumptions, we establish atom-wise local contraction of ITKrM with high probability towards such stable configurations up to an accuracy of order $\mathcal{O}(S/K)$, where S denotes the number of active atoms per signal. This limitation is caused by the mismatch of the dictionary sizes, which renders the thresholding step unpredictable in certain cases.

These results provide a first theoretical step towards understanding the behaviour of ITKrM under dictionary size mismatch and shed light on the robustness of dictionary learning algorithms in practice.

1. Introduction

In the field of signal processing and machine learning, dictionary learning plays an important role for sparse representations and analysis of data. The goal of dictionary learning is to find a dictionary $\Phi = (\phi_1, \dots, \phi_K) \in \mathbb{R}^{d\times K}$ for a number of given signals $y \in \mathbb{R}^d$, stored as columns of a matrix $Y = (y_1, \dots, y_N)$, such that

$$Y \approx \Phi X, \qquad \text{with} \qquad X \in \mathbb{R}^{K\times N} \text{ sparse.}$$

The coefficient matrix $X = (x_1, \dots, x_N)$ being sparse means that most entries of each coefficient vector x_n are zero. Over the years, a large variety of algorithms for dictionary learning have been proposed [11, 3, 10, 23, 15, 16, 17, 25, 21], accompanied by a growing body of theoretical results [12, 26, 4, 22, 13, 6, 5, 1, 2, 27, 28, 7, 24, 20, 19].

A common approach to dictionary learning is to formulate it as the optimisation problem

$$\operatorname{argmin}_{\Psi,X} \|Y - \Psi X\|_F^2 \qquad \text{s.t.} \qquad X \in \mathcal{S} \quad \text{and} \quad \|\psi_k\|_2 = 1 \text{ for all } k, \tag{1}$$

where $\mathcal{S}$ denotes a set imposing sparsity constraints on the coefficient matrix X, for example by assuming that each coefficient vector has at most $S \ll d$ dominant entries.

Since the optimisation problem in (1) is highly non-convex, many practical algorithms approach it via alternating optimisation schemes. These methods iteratively alternate between estimating sparse coefficients for a fixed dictionary and updating the dictionary for fixed coefficients. Prominent examples include MOD [9], K-SVD [3], ODL [17], and ITKrM [23]. Despite their strong empirical performance, analysing their convergence behaviour theoretically remains challenging due to the non-convex structure of the problem and the existence of many equivalent solutions induced by sign changes and permutations of the dictionary atoms.

Contribution: In this paper we study the convergence behaviour of the Iterative Thresholding and K-residual Means (ITKrM) algorithm in the practically relevant setting of dictionary size mismatch. Motivated by numerical experiments of [19], showing that dictionary learning algorithms often converge to stable configurations which do not coincide with the generating dictionary, we analyse the case where the learned dictionary contains one atom fewer than the generating dictionary. In this setting exact recovery is impossible and the algorithm compensates for the missing atom through merged structures.

We show that stable configurations of ITKrM consist of all generating atoms except that one pair of atoms is represented by a single merged atom of the form $\psi_i \pm \psi_j$ with $i \neq j$. Under standard sparsity and incoherence assumptions, we establish atom-wise local contraction of ITKrM with high probability towards such merged configurations. In particular, all remaining atoms contract towards their generating counterparts, while the merged atom contracts towards a normalized combination of two generating atoms. The achievable precision is limited by an error term of order $\mathcal{O}(S/K)$, caused by the ambiguity introduced through the dictionary size mismatch and the resulting instability of the thresholding step.

Our results provide a first theoretical step towards understanding the behaviour of ITKrM and alternating minimisation methods under misspecified dictionary sizes. They further demonstrate that stable fixed points of dictionary learning algorithms are not necessarily equivalent to the generating dictionary and help explain the characteristic of merged structures observed in practice.

Organisation: The remainder of this paper is organized as follows. In Section 2, we introduce the necessary notation and present the underlying model and assumptions. Section 3 contains the main theorem together with its proof. For the sake of readability, several auxiliary lemmas and technical results used in the proof are deferred to the Appendix.

2. Notation and Signal Model

The notation used in this paper is similar to the one in [19]. For the convenience of the reader we recall them here.

For a Matrix A we denote its transpose by A^* and its Moore-Penrose pseudoinverse by $A^\dagger$. For ease of notation we write for vectors $v \in \mathbb{R}^n$ the euclidean norm as $\|v\| := \|v\|_2$ and for matrices $A \in \mathbb{R}^{n\times m}$ we denote the operator norm $\|A\| := \|A\|_{2,2} = \max_{\|x\|_2=1} \|Ax\|_2$. Let $\Phi = (\phi_1, \dots, \phi_K) \in \mathbb{R}^{d\times K}$ be the generating dictionary with $\|\phi_i\|_2 = 1$ for all $i \in \mathbb{K} := \{1, \dots, K\}$. The coherence of a dictionary is defined as $\mu(\Phi) := \max_{i\neq j} |\langle \phi_i, \phi_j\rangle|$. For a support $I \subseteq \mathbb{K}$ with $|I| = S$, we write Φ_I for the corresponding subdictionary, that is, $\Phi_I = (\phi_{i_1}, \dots, \phi_{i_S})$ with $i_j \in I$. The orthogonal projection onto $\operatorname{span}(\Phi_I)$ is given by $P(\Phi_I) := \Phi_I\Phi_I^\dagger$ while the projection onto its orthogonal complement by $Q(\Phi_I) := \mathbb{I}_d - P(\Phi_I)$ where $\mathbb{I}_d$ denotes the identity on $\mathbb{R}^d$. We define the restricted isometry constant $\delta_I(\Phi)$ as the smallest number such that the spectrum of $\Phi_I^*\Phi_I$ is contained in $[1 - \delta_I(\Phi), 1 + \delta_I(\Phi)]$, and set $\delta_S(\Phi) := \max_{|I|\leq S} \delta_I(\Phi)$.

Pertubation model and distance. Since this paper covers the scenario of a dictionary size mismatch between the ground truth $\Phi \in \mathbb{R}d \times K$ and our current guess $\Psi \in \mathbb{R}^{d\times K-1}$, we assume that one atom of the current guess is a linear combination of two atoms of the ground truth Φ. This assumption comes from numeric experiments of [19]. Thus we model the first atom in the structure

$$\psi_1 = \alpha_1 \frac{\phi_1 + \phi_K}{\|\phi_1 + \phi_K\|} + \beta_1 \frac{\phi_1 - \phi_K}{\|\phi_1 - \phi_K\|} + \omega_1 z_1,$$

with $\langle \phi_1, z_1\rangle = \langle \phi_K, z_1\rangle = 0$. Meanwhile for $i \in \mathbb{K} \setminus \{1, K\}$ we use standard decomposition the

$$\psi_i = \alpha_i\phi_i + \omega_i z_i, \quad \text{with } \langle \phi_i, z_i\rangle = 0.$$

The coefficients satisfy the normalisation conditions $\alpha_i^2+\omega_i^2=1$ for $i\in\mathbb{K}\backslash\{1,K\}$ and $\alpha_1^2+\beta_1^2+\omega_1^2=1$. Throughout the paper we will often use that

$$\alpha_i = \begin{cases} \langle \phi_i, \psi_i \rangle, & i \in \mathbb{K} \setminus \{1,K\}, \\ \left\langle \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}, \psi_1 \right\rangle, & i \in \{1,K\}, \end{cases}$$

and define $\alpha_{\min} \leq \alpha_i \leq \alpha_{\max}$ for all $i \in \mathbb{K}$. We define the atom-wise ℓ_2-distance by

$$\varepsilon_i := \begin{cases} \|\phi_i - \psi_i\|, & i \in \mathbb{K} \setminus \{1,K\}, \\ \left\| \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|} - \psi_1 \right\|, & i \in \{1,K\}, \end{cases}$$

and set $\varepsilon := \max_{i\in\mathbb{K}} \varepsilon_i$. Thus we can relate the parameters to the perturbation size via

$$\alpha_i = 1 - \frac{\varepsilon_i^2}{2}, \quad \omega_i = \left(\varepsilon_i^2 - \frac{\varepsilon_i^4}{4}\right)^{1/2} \quad i \in \mathbb{K} \setminus \{1,K\}, \quad \text{and} \quad (\beta_1^2+\omega_1^2)^{1/2} = \left(\varepsilon_1^2 - \frac{\varepsilon_1^4}{4}\right)^{1/2}.$$

Sparse signal model The signal model correlates strongly with the signal models used in [19, 22, 23, 24]. A signal $y \in \mathbb{R}^d$ is generated by the dictionary $\Phi \in \mathbb{R}^{d\times K}$ according to

$$y = \Phi x = \sum_{i\in\mathbb{K}} \phi_i x_i,$$

where $x \in \mathbb{R}^K$ is a sparse coefficient sequence. For the coefficient sequence we first assume there is a measure ν_c on a subset $\mathcal{C}$ of positive, non increasing coefficient sequences with unit norm. Concretely this means $c \in \mathcal{C}$ satisfies $c(1) \geq c(2) \geq \ldots \geq c(K) \geq 0$ and $\|c\| = 1$. For $\sigma \in \{\pm 1\}^K$ being a sequence of independent Rademacher signs and a permutation p we model x as

$$x = x_{c,p,\sigma}, \quad \text{with} \quad x_{c,p,\sigma}(i) = \sigma(i)c(p(i)).$$

In order to have approximately S-sparse signals we enforce that the S largest coefficients are well balanced and much larger than the rest. Thus we define the support of a signal y as $I = p^{-1}(\mathbb{S})$. The energy of the coefficients outside the support we require to be relatively small. These conditions can be monitored by

$$\frac{c(1)}{c(S)} \leq \gamma_{\mathrm{dyn}}, \quad \frac{c(S+1)}{c(S)} \leq \gamma_{\mathrm{gap}} \quad \text{and} \quad \frac{\|c(\mathbb{S}^c)\|_2}{c(1)} \leq \gamma_{\mathrm{app}},$$

We refer to γ_{dyn} as the *dynamic range*, describing the ratio between the largest and smallest coefficients inside the support. The quantity γ_{gap} is called the *sparse gap* and measures the separation between coefficients inside and outside the support. The last parameter γ_{app} is the *relative approximation error*, controlling the energy of coefficients outside the support.

Algorithm 1 ITKrM (one iteration)

Input: Signals $\{y_n\}_{n=1}^N \subset \mathbb{R}^d$, dictionary $\Psi = (\psi_1, \ldots, \psi_{K-1})$, sparsity level S
Initialize $\tilde{\Psi} = 0, \hat{X} = 0$
for $n = 1, \ldots, N$ **do**
  $\hat{I}_n = \operatorname{argmax}_{I:|I|=S} \|\Psi_I^* y_n\|_1$ (thresholding)
  $\hat{x}_n(\hat{I}_n) = \Psi_{\hat{I}_n}^\dagger y_n$ (coefficient estimation)
  $a_n = y_n - \Psi_{\hat{I}_n} \hat{x}_n$ (residual)
  for $k \in \hat{I}_n$ **do**
    $\tilde{\psi}_k \leftarrow \tilde{\psi}_k + \frac{1}{N}\big(a_n + \psi_k\langle\psi_k, y_n\rangle\big)\ \mathrm{sign}(\langle\psi_k, y_n\rangle)$ (dictionary update)
  end for
end for
$\hat{\Psi} \leftarrow (\tilde{\psi}_1/\|\tilde{\psi}_1\|_2, \ldots, \tilde{\psi}_{K-1}/\|\tilde{\psi}_{K-1}\|_2)$
Output: $\hat{\Psi}$

3. Main Result

We now state our main result.

Theorem 1 *Assume the signals follow the model in 2 for coefficients with gap* $c(S+1)/c(S) \leq \gamma_{gap}$*, dynamic sparse range* $c(1)/c(S) \leq \gamma_{dyn}$*, and relative approximation error* $\|c(\mathcal{S}^c)\|_2/c(1) \leq \gamma_{app} \leq \frac{12}{7}\sqrt{\log K}$*. If the cross Gram matrix* $\Phi^*\Psi$ *is diagonally dominant in the sense that*

$$\begin{aligned}\min\left\{\min_{k\in\mathbb{K}\setminus\{1,K\}} |\langle\psi_k,\phi_k\rangle|, \left|\left\langle\psi_1, \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\right\rangle\right|\right\} \\ \geq \max\Bigg\{\frac{8}{3}\sqrt{2(1+\langle\phi_1,\phi_K\rangle)}\cdot\gamma_{gap}\cdot\left|\left\langle\psi_1, \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\right\rangle\right|, \\ \frac{8}{3}\gamma_{gap}\cdot\max_{k\notin\{1,K\}}|\langle\psi_k,\phi_k\rangle|, \\ 58\gamma_{dyn}\cdot\mu(\Phi,\Psi)\cdot\log K, \\ \frac{50}{3}\gamma_{dyn}\cdot\sqrt{\frac{S\|\Phi\|_{2,2}^2\log K}{K-S}}\Bigg\}\end{aligned} \tag{2}$$

and additionally

$$\begin{aligned}\left|\left\langle\psi_1, \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\right\rangle\right| \geq \max\Bigg\{\left|\left\langle\psi_1, \frac{\phi_1-\phi_K}{\|\phi_1-\phi_K\|}\right\rangle\right|, \\ \frac{2}{\sqrt{2(1+\langle\phi_1,\phi_K\rangle)}\cdot(1-\gamma_{gap})-\frac{20}{16}}\cdot\gamma_{gap}\cdot\max_{k\notin\{1,K\}}|\langle\psi_k,\phi_k\rangle| \\ -\frac{\sqrt{2(1+\langle\phi_1,\phi_K\rangle)}}{2}c(S)\left|\left\langle\psi_1, \frac{\phi_1-\phi_K}{\|\phi_1-\phi_K\|}\right\rangle\right|\Bigg\},\end{aligned} \tag{3}$$

as well as that

$$\mu(\Psi) \leq \frac{1}{20\log K} \quad \textit{and} \quad \|\Psi\|_{2,2}^2 \leq \frac{K}{134e^2 S\log K} - 1, \tag{4}$$

then for $i\mathbb{K}\setminus\{1,K\}$ *the updated and normalised atoms* $\bar{\psi}_i$ *- the output of ITKrM - satisfies*

$$\left\|\frac{\bar{\psi}_i}{\|\bar{\psi}_i\|} - \phi_i\right\|_2 \leq 0.708\varepsilon + err \quad \textit{and}$$
$$\left\|\frac{\bar{\psi}_1}{\|\bar{\psi}_1\|} - \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\right\|_2 \leq 0.708\varepsilon + err_{1K},$$

with $err \leq \frac{\sqrt{2}}{0.95}\cdot\frac{\|\Phi\|_{2,2}}{c(S)}\cdot\frac{S^2+16+16/S}{K^2}$ *and* $err_{1K} \leq \frac{2\sqrt{1+\langle\phi_1,\phi_K\rangle}}{0.95}\cdot\frac{\|\Phi\|_{2,2}}{c(S)}\cdot\frac{S}{K}\cdot\left(\frac{2}{K^2}+\frac{12}{KS}+\frac{1}{2}\right)$*, except with probability*

$$\begin{aligned}28(K-2)\exp\left(-N\left(\frac{72(\|\Phi\|_{2,2}^2+c(1))^2K^2}{\gamma_{1,S}^2\varepsilon^2}+\frac{96(S+1)+6S^3}{K^2\gamma_{1,S}\varepsilon}+2\right)^{-1}\right) \\ +28\exp\left(-N\left(\frac{36(\|\Phi\|_{2,2}^2+2c(1))^2K^2}{\gamma_{1,S}^2\varepsilon^2}+\frac{69(S+1)+5S^2}{2K\gamma_{1,S}\varepsilon}+2\right)^{-1}\right) \\ +K\exp\left(-\frac{N\gamma_{1,S}^2}{800K\left(1+\frac{S\|\Phi\|_{2,2}^2}{K}+\gamma_{1,S}\|\Phi\|_{2,2}/60\right)}\right).\end{aligned}$$

The proof decomposes into several auxiliary estimates. To improve readability, we state the main contraction argument first and defer technical lemmas and concentration bounds to the appendix. We begin by restating the matrix resp. vector Bernstein theorem, which is fundamental for the proof of the theorem above.

Theorem 2 (Matrix resp. Bernstein [30], [18]) *Consider a sequence $\{Y_n\}_{n=1}^N$ of independent, random matrices (resp. vectors) with dimension $d \times K$ (resp. d). Assume that each random matrix (resp. vector) satisfies*

$$\|Y_n\| \le R \quad a.s. \qquad and \qquad \|\mathbb{E}[Y_n]\| \le m.$$

Then, for all $t > 0$,

$$\mathbb{P}\left(\left\|\frac{1}{N}\sum_{n=1}^{N} Y_n\right\| \ge m + t\right) \le \kappa \exp\left(\frac{-Nt^2}{2R^2 + (R+m)t}\right), \tag{32}$$

where $\kappa = d + K$ for the matrix Bernstein inequality and $\kappa = 28$ for the vector Bernstein inequality.

Proof of Theorem 1

Throughout the proof we use the abbriviations $B := \|\Phi\|_{2,2}^2$ and $\bar{B} := \|\Psi\|_{2,2}^2$. The idea of the proof is to apply vector Bernstein by rewriting the update step of ITKrM as a sum of independent random vectors $\hat{Y}_n$, which we define as

$$\hat{Y}_n := \Big[\mathbb{I} - P(\Psi_{\hat{I}_n}) + P(\psi_i)\Big] \cdot y_n \cdot \operatorname{sign}(\langle \phi_i, y_n\rangle) \cdot \mathbb{1}_{\hat{I}_n}(i) - c_n(i)\phi_i \cdot \mathbb{1}_{I_n}(i),$$

with $i \in \hat{I}_n \setminus \{1, K\}$. Since $\psi_1 = \alpha_1 \cdot \frac{\phi_1 + \phi_K}{\|\phi_1 + \phi_K\|} + \beta_1 \cdot \frac{\phi_1 - \phi_K}{\|\phi_1 - \phi_K\|} + \omega_1 \cdot z_1$ is a special case, we cover the contraction of $\bar{\psi}_1$ separately in 3. We can see that $\hat{Y}_n$ only depends on the signal y_n and thus the vectors $\hat{Y}_n$ are independent of each other. Further we get

$$\frac{1}{N}\sum_{n=1}^{N} \hat{Y}_n = \bar{\psi}_i - s_i \cdot \phi_i,$$

with $s_i = \frac{1}{N}\sum_n |\langle \phi_i, y_n\rangle| \cdot \mathbb{1}_{I_n}(i)$. This allows us to use the Bernstein inequality. We bound the norm using $\|y_n\|^2 = \|\Phi x\|^2 \le \|\Phi\|_{2,2}^2 \cdot \|x\|^2 = B \cdot \|c\|^2 = B$ which yields

$$\|\hat{Y}_n\| \le \|\mathbb{I} + P(\Psi_{\hat{I}_n}) + P(\psi_i)\| \cdot \|y_n\| + \|c_n(i)\phi_i\| \le \sqrt{B} + c_n(1) =: R.$$

The estimation of $\|\mathbb{E}[\hat{Y}_n]\|$ follows a more complex approach. Thus we define another random vector Y_n for each n, for which the estimated support $\hat{I}_n$ and $\operatorname{sign}(\langle \phi_i, y_n\rangle)$ is replaced with the correct support I_n and sign $\sigma(i)_n$, yielding

$$Y_n := ([\mathbb{I} - P(\Psi_{I_n}) + P(\psi_i)] \cdot y_n \cdot \sigma_n(i) - c(i)\phi_i) \cdot \mathbb{1}_{I_n}(i)$$

Out of convenience, we drop the index n. Comparing the random variables $\hat{Y}$ and Y we notice that they only differ, when thresholding fails to recover the correct support I or the correct sign $\sigma(i)$, which can be written in the following set:

$$\mathcal{R} := \Big\{(I, \sigma, c) : \Big[\Big(\hat{I} \ne I \vee \operatorname{sign}(\langle \psi_i, y\rangle) \ne \sigma(i)\Big) \wedge i \in I\Big] \vee \Big[i \in \hat{I} \wedge i \notin I\Big]\Big\}.$$

Using that $\|Y\| \le \sqrt{B} + c(1)$ we get

$$\|\mathbb{E}[\hat{Y}]\| \le \|\mathbb{E}[\hat{Y} - Y]\| + \|\mathbb{E}[Y]\| \le \|\hat{Y} - Y\| \cdot \mathbb{P}(\mathcal{R}) + \|\mathbb{E}[Y]\| \le 2\sqrt{B} \cdot \mathbb{P}(\mathcal{R}) + \|\mathbb{E}[Y]\|. \tag{5}$$

Using the fact that the product of two Rademacher random variables is zero in expectation, meaning $\mathbb{E}[\sigma(i)\sigma(j)] = 0$, we obtain

$$
\begin{aligned}
\mathbb{E}[Y] &= \mathbb{E}\left[\left([\mathbb{I} - P(\Psi_I) + P(\psi_i)] \cdot y \cdot \sigma(i) - c(i)\phi_i\right) \cdot \mathbb{1}_I(i)\right] \\
&= \mathbb{E}\left[\left([\mathbb{I} - P(\Psi_I) + P(\psi_i)] \cdot \left[c(i)\phi_i + \sum_{j\neq i} \sigma(i)\sigma(j)c(j)\phi_j\right] - c(i)\phi_i\right) \cdot \mathbb{1}_I(i)\right] \\
&= \mathbb{E}\left[c(i) \cdot (P(\psi_i)\phi_i - P(\Psi_I)\phi_i) \cdot \mathbb{1}_I(i)\right] \\
&= \frac{\gamma_{1,S}}{K} \cdot \binom{K-1}{S-1}^{-1} \sum_{|I|=S, i\in I} P(\psi_i)\phi_i - P(\Psi_I)\phi_i,
\end{aligned}
\tag{6}
$$

Extending ϕ_i with the orthogonal decomposition $\phi_i = [P(\psi_i) + Q(\psi_i)]\,\phi_i$ and using that $P(\psi_i)\,Q(\psi_i) = 0$, we get

$$
\mathbb{E}[Y] = \frac{\gamma_{1,S}}{K}\binom{K-1}{S-1}^{-1} \sum_{|I|=S, i\in I} -P(\Psi_I)\,Q(\psi_i)\,\phi_i. \tag{7}
$$

Since the perturbed dictionary Ψ is well-conditioned and incoherent, for most I the subdictionary Ψ_I will be a quasi isometry and $P(\Psi_I) \approx \Psi_I\Psi_I^*$. We therefore expand the expectation above, using the abbreviation $p_{K,S} = \binom{K-1}{S-1}^{-1}$, as

$$
\begin{aligned}
\frac{K}{\gamma_{1,S}}\mathbb{E}[Y] =& p_{K,S}\left(\sum_{|I|=S, i\in I} [\Psi_I\Psi_I^* - P(\Psi_I)]\,Q(\psi_i)\,\phi_i - \sum_{|I|=S, i\in I} \Psi_{I\setminus i}\Psi_{I\setminus i}^* Q(\psi_i)\,\phi_i\right) \\
=& p_{K,S}\left(\sum_{|I|=S, i\in I} [\Psi_I\Psi_I^* - P(\Psi_I)]\,Q(\psi_i)\,\phi_i - \binom{K-2}{S-2}\sum_{j\neq i} \psi_j\psi_j^* Q(\psi_i)\,\phi_i\right) \\
=& p_{K,S}\sum_{|I|=S, i\in I} [\Psi_I\Psi_I^* - P(\Psi_I)]\,Q(\psi_i)\,\phi_i - \frac{S-1}{K-1}\left(\Psi\Psi^* - \psi_i\psi_i^*\right) Q(\psi_i)\,\phi_i \\
=& p_{K,S}\sum_{\substack{|I|=S, i\in I \\ \delta(\Psi_I)\leq\delta_0}} [\Psi_I\Psi_I^* - P(\Psi_I)]\,Q(\psi_i)\,\phi_i \\
&+ p_{K,S}\sum_{\substack{|I|=S, i\in I \\ \delta(\Psi_I)>\delta_0}} [\Psi_I\Psi_I^* - P(\Psi_I)]\,Q(\psi_i)\,\phi_i - \frac{S-1}{K-1}\Psi\Psi^* Q(\psi_i)\,\phi_i.
\end{aligned}
\tag{8}
$$

Since for $\psi_i = \alpha_i\phi_i + \omega_i z_i$ we have $\|Q(\psi_i)\,\phi_i\|_2 = \omega_i \leq \varepsilon$ whenever $i \in \mathbb{K} \setminus \{1, K\}$, we can bound the norm of the expectation above as

$$
\|\mathbb{E}[Y]\|_2 \leq \frac{\gamma_{1,S}}{K}\left[\delta_0 + \mathbb{P}\left(\delta(\Psi_I) > \delta_0 \mid |I| = S, i \in I\right) \cdot (\bar{B}+1) + \frac{(S-1)\bar{B}}{K-1}\right]\varepsilon.
$$

To estimate the probability of a subdictionary being ill-conditioned we use Chretien and Darses's results on the conditioning of random subdictionaries, which are slightly cleaner and thus easier to handle than the original results by Tropp, [29]. Theorem 3.1 of [8] reformulated for our purposes and applied to Ψ states that

$$
\mathbb{P}\left(\delta(\Psi_I) > \delta_0 \mid |I| = S\right) \leq 216K \exp\left(-\min\left\{\frac{\delta_0}{2\mu(\Psi)}, \frac{\delta_0^2 K}{4e^2 S\bar{B}}\right\}\right).
$$

Together with the union bound,

$$\mathbb{P}\left(\delta\left(\Psi_I\right) > \delta_0 \mid |I| = S, i \in I\right) \leq \frac{K}{S} \cdot \mathbb{P}\left(\delta\left(\Psi_I\right) > \delta_0 \mid |I| = S\right),$$

this leads to

$$\|\mathbb{E}[Y]\|_2 \leq \frac{\gamma_{1,S}}{K}\left[\delta_0 + \frac{216K^2(\bar{B}+1)}{S}\exp\left(-\min\left\{\frac{\delta_0}{2\mu(\Psi)}, \frac{\delta_0^2 K}{4e^2 S\bar{B}}\right\}\right) + \frac{(S-1)\bar{B}}{K-2}\right]\varepsilon.$$

We set $\delta_0 = 3/10$ and recall the assumptions of Theorem 3.1 of [8], which are $\bar{B} \leq \frac{K}{134e^2 S\log K} - 1$ and $\mu(\Psi) \leq \frac{1}{20\log K}$, to get a bound of $\mathbb{E}[\hat{Y}]$ from (10):

$$\|\mathbb{E}[\hat{Y}]\| \leq 0.309 \cdot \frac{\gamma_{1,S}}{K} \cdot \varepsilon + 2\sqrt{B} \cdot \mathbb{P}(\mathcal{R}) =: m.$$

We are now able to apply vector Bernstein with $t = \frac{1}{6} \cdot \frac{\gamma_{1,S}}{K} \cdot \varepsilon$, which gives us with high probability $p_1 \leq 28\exp\left(-N\left(\frac{72(B+c(1))^2K^2}{\gamma_{1,S}^2\varepsilon^2} + \frac{96(S+1)+6S^3}{K^2\gamma_{1,S}\varepsilon} + 1.854\right)^{-1}\right)$ that

$$\|\bar{\psi}_i - s_i\phi_i\| \leq m + t \leq \left(0.475 \cdot \varepsilon + \frac{2K\sqrt{B}}{\gamma_{1,S}} \cdot \mathbb{P}(\mathcal{R})\right) \cdot \frac{\gamma_{1,S}}{K}, \tag{9}$$

for $i \in \mathbb{K} \setminus \{1, K\}$ and $s_i = \frac{1}{N}\sum_n |\langle\phi_i, y_n\rangle| \cdot \mathbb{1}_{I_n}(i)$. From Lemma 5 we know for $\hat{t} = 0.05$ that $s_i \geq (1-\hat{t}) \cdot \gamma_{1,S}/K = 0.95 \cdot \gamma_{1,S}/K$ with probability $p_2 \leq \exp\left(-\frac{N\gamma_{1,S}^2}{800K\left(1+\frac{SB}{K}+\gamma_{1,S}\sqrt{B}/60\right)}\right)$. We normalise $\bar{\psi}_i$ and apply Lemma 6, yielding

$$\begin{aligned}\left\|\frac{\bar{\psi}_i}{\|\bar{\psi}_i\|} - \phi_i\right\|_2^2 &\leq 2 - 2\sqrt{1 - \frac{(0.475 \cdot \varepsilon + 2K\sqrt{B} \cdot \mathbb{P}(\mathcal{R})/\gamma_{1,S})^2}{0.95^2}} \\ &\leq \frac{2 \cdot (0.475 \cdot \varepsilon + 2K\sqrt{B} \cdot \mathbb{P}(\mathcal{R})/\gamma_{1,S})^2}{0.95^2} \leq (0.708\varepsilon + \mathrm{err})^2,\end{aligned}$$

with probability $(K-2)(p_1+p_2)$ and $\mathrm{err} = \frac{\sqrt{8}}{0.95}\frac{\sqrt{B}}{\gamma_{1,S}}K \cdot \mathbb{P}(\mathcal{R})$. Using Lemma 3 on $\mathbb{P}(\mathcal{R})$ and the trivial bound $\gamma_{1,S} \geq Sc(S)$ we obtain $\mathrm{err} \leq \frac{\sqrt{2}}{0.95} \cdot \frac{\sqrt{B}}{c(S)} \cdot \frac{S^2+16+16/S}{K^2}$.

Case ψ_1: The idea of the proof is to apply vector Bernstein by rewriting $\bar{\psi}_1 - \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}$ as a sum of independent random vectors $\hat{Y}_n$, which we define as

$$\begin{aligned}\hat{Y}_n := \left[\mathbb{I} - P(\Psi_{\hat{I}_n}) + P(\psi_1)\right] \cdot y_n \cdot \operatorname{sign}\left(\left\langle \phi_1 \cdot \mathbb{1}_{\hat{I}_n}(1) \;+\; \phi_K \cdot \mathbb{1}_{\hat{I}_n}(K), y_n \right\rangle\right) \\ - c_n(1)\phi_1 \cdot \mathbb{1}_{I_n}(1) - c_n(K)\phi_K \cdot \mathbb{1}_{I_n}(K).\end{aligned}$$

Again we can see that $\hat{Y}_n$ only depends on the signal y_n and thus the vectors $\hat{Y}_n$ are independent of each other. Further we get

$$\frac{1}{N}\sum_{n=1}^{N}\hat{Y}_n = \bar{\psi}_1 - (s_1\phi_1 + s_K\phi_K),$$

with $s_1 = \frac{1}{N}\sum_n |\langle\phi_1, y_n\rangle| \cdot \mathbb{1}_{I_n}(1)$ and $s_K = \frac{1}{N}\sum_n |\langle\phi_K, y_n\rangle| \cdot \mathbb{1}_{I_n}(K)$. This allows us to use the Bernstein inequality. Again we drop the index n and bound the operator norm using $\|y\|^2 \leq B$ which yields

$$\|\hat{Y}\| \leq \|\mathbb{I} + P(\Psi_{\hat{I}}) + P(\psi_1)\| \cdot \|y\| + \|c(1)\phi_1 + c(K)\phi_K\| \leq \sqrt{B} + \sqrt{2(1+\langle\phi_1, \phi_K\rangle} \cdot c(1).$$

The estimation of $\|\mathbb{E}[\hat{Y}_n]\|$ follows a more complex approach, similar as in case $i \in \mathbb{K}\backslash\{1, K\}$. Thus we define another random vector Y, for which the estimated support $\hat{I}$ and sign $\left(\left\langle \phi_1 \cdot \mathbb{1}_{\hat{I}_n}(1)\phi_K \cdot \mathbb{1}_{\hat{I}_n}(K), y_n\right\rangle\right)$ is replaced with the correct support I and sign $\sigma(i)$ for $i \in \{1, K\}$, yielding

$$Y := [\mathbb{I} - P(\Psi_I) + P(\psi_1)] \cdot y \cdot \operatorname{sign}(\sigma(1) \cdot \mathbb{1}_{I_n}(1) + \sigma(K) \cdot \mathbb{1}_{I_n}(K)) - c(1)\phi_1 \cdot \mathbb{1}_I(1) - c(K)\phi_K \cdot \mathbb{1}_I(K)$$

Next we recall that thresholding can only recover the correct support I, if $K \notin I$, since Ψ has one less atom than Φ and thus covers the missing atom ϕ_K with the double atom $\psi_1 = \alpha_1 \cdot \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|} + \beta_1 \cdot \frac{\phi_1-\phi_K}{\|\phi_1-\phi_K\|} + \omega_1 \cdot z_1$. Hence we say that thresholding also recovers the correct support I with $K \in I$ and $1 \notin I$, if $\hat{I} = I \setminus \{K\} \cup \{1\} =: I_K$. Whenever $1, K \in I$ thresholding cannot recover the correct support, since $|\hat{I}|$ is enforced to have S elements and hence has to choose some $i \in I^c$. Thus we define the best recovery of the support as $\hat{I} =: I_{1K}$, if $\hat{I} \in \{\tilde{I} \in \mathbb{K}^S : \tilde{I} = I \setminus \{K\} \cup \{i\}$ for some $i \in I^c\}$. Comparing the random variables $\hat{Y}$ and Y we notice that they only differ, when thresholding fails to recover the correct support I and the correct sign $\sigma(i)$ for $i \in \{1, K\}$. These cases can be written in the following set:

$$\begin{aligned}\mathcal{R}_{1K} := \Big\{(I, \sigma, c) : &\Big[\Big(\hat{I} \neq I \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(1)\Big) \wedge (1 \in I \wedge K \notin I)\Big] \\ &\vee \Big[\Big(\hat{I} \neq I_K \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(K)\Big) \wedge (1 \notin I \wedge K \in I)\Big] \\ &\vee \Big[\Big(\hat{I} \neq I_{1K} \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(1) \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(K)\Big) \wedge (1 \in I \wedge K \in I)\Big] \\ &\vee \Big[1 \in \hat{I} \wedge 1, K \notin I\Big]\Big\}.\end{aligned}$$

Using that $\|Y\| \leq \sqrt{B} + \sqrt{2+2\mu} \cdot c(1)$ leads to

$$\|\mathbb{E}[\hat{Y}]\| \leq \|\mathbb{E}[\hat{Y} - Y]\| + \|\mathbb{E}[Y]\| \leq \|\hat{Y} - Y\| \cdot \mathbb{P}(\mathcal{R}_{1K}) + \|\mathbb{E}[Y]\| \leq 2\sqrt{B} \cdot \frac{S^2}{K^2} \cdot \mathbb{P}(\mathcal{R}_{1K}) + \|\mathbb{E}[Y]\|. \quad (10)$$

We continue by defining

$$\begin{aligned} Y^1 &:= \Big(\Big[\mathbb{I} - \Psi_I \Psi_I^\dagger + \psi_1\psi_1^*\Big] \cdot y \cdot \sigma(1) - c(1)\phi_1\Big) \cdot \mathbb{1}_I(1) \cdot \mathbb{1}_{I^c}(K) \\ Y^K &:= \Big(\Big[\mathbb{I} - \Psi_{I_K} \Psi_{I_K}^\dagger + \psi_1\psi_1^*\Big] \cdot y \cdot \sigma(K) - c(K)\phi_K\Big) \cdot \mathbb{1}_I(K) \cdot \mathbb{1}_{I^c}(1) \\ Y^{1K} &:= \Big(\Big[\mathbb{I} - \Psi_{I_{1K}} \Psi_{I_{1K}}^\dagger + \psi_1\psi_1^*\Big] \cdot y \cdot \sigma(1) - c(1)\phi_1 - c(K)\phi_K\Big) \cdot \mathbb{1}_I(\{1, d\}) \cdot \mathbb{1}(\sigma(1) = \sigma(K)), \end{aligned}$$

and hence get

$$\mathbb{E}[Y] = \mathbb{E}[Y^1 + Y^K + Y^{1K}]. \quad (11)$$

The next steps are easier to understand, if we observe each summand separately, before we add them again. We recall that $\gamma_{1,S} = \mathbb{E}[c(1) + \ldots + c(S)]$ and hence $\mathbb{E}[c(1) \cdot \mathbb{1}_I(1) \cdot \mathbb{1}_{I^c}(K)] = \frac{\gamma_{1,S} \cdot (K-S)}{K(K-1)}$. We rewrite the first summand Y^1 as in (6) and use the orthogonal decomposition as in (7), yielding

$$\begin{aligned} \mathbb{E}[Y^1] &= \frac{\gamma_{1,S}(K-S)}{K(K-1)} \cdot \binom{K-2}{S-1}^{-1} \sum_{|I|=S, 1 \in I, K \notin I} P(\psi_1)\phi_1 - P(\Psi_I)\phi_1 \\ &= \frac{\gamma_{1,S}(K-S)}{K(K-1)} \cdot \binom{K-2}{S-1}^{-1} \sum_{|I|=S, 1 \in I, K \notin I} -P(\Psi_I)\, Q(\psi_1)\, \phi_1. \quad (12) \end{aligned}$$

We make the same steps with $\mathbb{E}[Y^K]$, yielding

$$\mathbb{E}[Y^K] = \frac{\gamma_{1,S}(K-S)}{K(K-1)} \cdot \binom{K-2}{S-1}^{-1} \sum_{|I|=S, K\in I, 1\notin I} -P\left(\Psi_{I_K}\right) Q\left(\psi_1\right) \phi_K. \tag{13}$$

We note that, because of I_K in (13), we sum over the same supports as in (12). Thus we can combine $\mathbb{E}[Y^1]$ and $\mathbb{E}[Y^K]$ in the following manner

$$\mathbb{E}[Y^1 + Y^K] = \frac{\gamma_{1,S}(K-S)}{K(K-1)} \cdot \binom{K-2}{S-1}^{-1} \sum_{|I|=S, 1\in I, K\notin I} -P\left(\Psi_{I_K}\right) Q\left(\psi_1\right) \left(\phi_1 + \phi_K\right). \tag{14}$$

We perform the same steps as above with the last summand Y^{1K}, yielding

$$\mathbb{E}[Y^{1K}] = \frac{\gamma_{1,S}(S-1)}{K(K-1)} \cdot \binom{K-2}{S-1}^{-1} \sum_{|I_{1K}|=S, 1\in I_{1K}} -P\left(\Psi_{I_{1K}}\right) Q\left(\psi_1\right) \left(\phi_1 + \phi_K\right),$$

Again we note that, because of I_{1K}, we sum over the same projections as in (14), which gives us

$$\begin{aligned}
\mathbb{E}[Y] &= \mathbb{E}[Y^1 + Y^K + Y^{1K}] \\
&= \frac{\gamma_{1,S}}{K} \cdot \binom{K-2}{S-1}^{-1} \sum_{|I_{1K}|=S, 1\in I_{1K}} -P\left(\Psi_{I_{1K}}\right) Q\left(\psi_1\right) \left(\phi_1 + \phi_K\right)
\end{aligned}$$

We continue as in (8) and exploit that $P\left(\Psi_{I_{1K}}\right) \approx \Psi_{I_{1K}} \Psi_{I_{1K}}^*$ yielding

$$\begin{aligned}
\frac{K}{\gamma_{1,S}} \mathbb{E}[Y] =& \binom{K-2}{S-1}^{-1} \sum_{\substack{|I_{1K}|=S, 1\in I_{1K} \\ \delta\left(\Psi_{I_{1K}}\right)\le \delta_0}} \left[\Psi_{I_{1K}} \Psi_{I_{1K}}^* - P\left(\Psi_{I_{1K}}\right)\right] Q\left(\psi_1\right) \left(\phi_1 + \phi_K\right) \\
&+ \binom{K-2}{S-1}^{-1} \sum_{\substack{|I_{1K}|=S, 1\in I_{1K} \\ \delta\left(\Psi_{I_{1K}}\right)> \delta_0}} \left[\Psi_{I_{1K}} \Psi_{I_{1K}}^* - P\left(\Psi_{I_{1K}}\right)\right] Q\left(\psi_1\right) \left(\phi_1 + \phi_K\right) \\
&- \frac{S-1}{K-2} \Psi \Psi^* Q\left(\psi_1\right) \left(\phi_1 + \phi_K\right).
\end{aligned}$$

We recall that $n_{1+K} = \|\phi_1 + \phi_K\|$ and get $\|Q\left(\psi_1\right)\left(\phi_1 + \phi_K\right)\|_2 = n_{1+K}\sqrt{1-\alpha_1^2} = n_{1+K}\sqrt{\beta_1^2 + \omega_1^2} \le n_{1+K}\varepsilon$, since $\psi_1 = \alpha_1 \cdot \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|} + \beta_1 \cdot \frac{\phi_1-\phi_K}{\|\phi_1-\phi_K\|} + \omega_1 \cdot z_1$. We further bound the norm of the expectation above as in step 1, yielding

$$\begin{aligned}
\|\mathbb{E}[Y]\|_2 &\le \frac{\gamma_{1,S} \cdot n_{1+K}}{K} \left[\delta_0 + \mathbb{P}\left(\delta\left(\Psi_{I_{1K}}\right) > \delta_0 \mid |I_{1K}| = S, 1 \in I_{1K}\right) \cdot (\bar{B}+1) + \frac{(S-1)\bar{B}}{K-2}\right] \varepsilon \\
&\le \frac{\gamma_{1,S} \cdot n_{1+K}}{K} \left[\delta_0 + \frac{216K^2(\bar{B}+1)}{S} \exp\left(-\min\left\{\frac{\delta_0}{2\mu(\Psi)}, \frac{\delta_0^2 K}{4e^2 S\bar{B}}\right\}\right) + \frac{(S-1)\bar{B}}{K-2}\right] \varepsilon.
\end{aligned}$$

Lastly, we can put all bounds together. We choose $\delta_0 = 3/10$ along with the assumptions of Theorem 3.1 of [8], which are $\bar{B} \le \frac{K}{134e^2 S\log K} - 1$ and $\mu(\Psi) \le \frac{1}{20\log K}$ and obtain

$$\|\mathbb{E}[Y]\| \le \frac{0.309 n_{1+K}\gamma_{1,S}}{K} \cdot \varepsilon + \sqrt{B} \cdot \mathbb{P}(\mathcal{R}_{1K}) =: m.$$

We are now able to apply vector Bernstein with $t = \frac{1}{6} \cdot \frac{n_{1+K}\gamma_{1,S}}{K} \cdot \varepsilon$, which gives us with probability $p_1 \leq 28 \exp\left(-N\left(\frac{72(B+\sqrt{2(1+\langle\phi_1,\phi_K\rangle)}c(1))^2K^2}{2(1+\langle\phi_1,\phi_K\rangle)\gamma_{1,S}^2\varepsilon^2} + \frac{96(S+1)+6S^2}{K\sqrt{2(1+\langle\phi_1,\phi_K\rangle)}\gamma_{1,S}\varepsilon} \cdot \left(\frac{2}{K^2} + \frac{12}{KS} + \frac{1}{2}\right) + 1.854\right)^{-1}\right)$ that

$$\|\bar{\psi}_i - (s_1\phi_1 + s_K\phi_K)\| \leq m + t \leq 0.475 \cdot \frac{n_{1+K}\gamma_{1,S}}{K} \cdot \varepsilon + \sqrt{B} \cdot \mathbb{P}(\mathcal{R}_{1K}), \tag{15}$$

for $s_1 = \frac{1}{N}\sum_n |\langle\phi_1, y_n\rangle| \cdot \mathbb{1}_{I_n}(1)$ and $s_K = \frac{1}{N}\sum_n |\langle\phi_K, y_n\rangle| \cdot \mathbb{1}_{I_n}(K)$. From Lemma 5 we know for $\hat{t} = 0.05$ that $\min\{s_1, s_K\} \geq (1-\hat{t}) \cdot \gamma_{1,S}/K = 0.95 \cdot \gamma_{1,S}/K =: s_{1K}$ except with probability $p_2 \leq 2\exp\left(-\frac{N\gamma_{1,S}^2}{800K\left(1+\frac{SB}{K}+\gamma_{1,S}\sqrt{B}/60\right)}\right)$. We apply Lemma 6 on $\|\bar{\psi}_1 - (n_{1+K} \cdot s_{1K})\frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\|$, which yields

$$\begin{aligned}
\left\|\frac{\bar{\psi}_1}{\|\bar{\psi}_1\|} - \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\right\|_2^2 &\leq 2 - 2\sqrt{1 - \frac{(0.475\varepsilon + K\sqrt{B}\mathbb{P}(\mathcal{R}_{1K})/(n_{1+K}\gamma_{1,S}))^2}{0.95^2}} \\
&\leq \frac{2(0.475\varepsilon + K\sqrt{B}\mathbb{P}(\mathcal{R}_{1K})/(n_{1+K}\gamma_{1,S}))^2}{0.95^2} \\
&\leq \left(0.708\varepsilon + \frac{\sqrt{2}\cdot K\sqrt{B}\mathbb{P}(\mathcal{R}_{1K})}{0.95 \cdot n_{1+K}\gamma_{1,S}}\right)^2 \\
&\leq (0.708\varepsilon + \mathrm{err}_{1K})^2,
\end{aligned}$$

except with probability $p_1 + p_2$, where $\mathrm{err}_{1K} = \frac{\sqrt{2}\cdot K\sqrt{B}\cdot\mathbb{P}(\mathcal{R}_{1K})}{0.95\cdot n_{1+K}\gamma_{1,S}}$. Using Lemma 3 on $\mathbb{P}(\mathcal{R}_{1K})$ and the trivial bound $\gamma_{1,S} \geq Sc(S)$ we obtain $\mathrm{err}_{1K} \leq \frac{2\sqrt{1+\langle\phi_1,\phi_K\rangle}}{0.95} \cdot \frac{\sqrt{B}}{c(S)} \cdot \frac{S}{K} \cdot \left(\frac{2}{K^2} + \frac{12}{KS} + \frac{1}{2}\right)$.

4. Appendix

To keep the presentation of the main proof concise, we defer several auxiliary estimates to the appendix. The following lemma gives a bound of the probability that thresholding fails to recover the correct support under our setting, while the other lemmas were proven in [19] and [24].

Lemma 3 *Assume the signals follow the model in 2 for coefficients with gap $c(S+1)/c(S) \leq \gamma_{gap}$, dynamic sparse range $c(1)/c(S) \leq \gamma_{dyn}$, relative approximation error $\|c(\mathcal{S}^c)\|_2/c(1) \leq \gamma_{app} \leq \frac{12}{7}\sqrt{\log K}$. We define $I_K := I\backslash\{K\}\cup\{1\}$ and $I_{1K} := I\backslash\{K\}\cup\{i\}$ for some $i \in I^c$, and $\langle\phi_1, \phi_K\rangle \geq 0$. If the cross Gram matrix $\Phi^*\Psi$ is diagonally dominant in the sense that*

$$\begin{aligned}
\min\left\{\min_{k\in\mathbb{K}\backslash\{1,K\}} |\langle\psi_k, \phi_k\rangle|, \left|\left\langle\psi_1, \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\right\rangle\right|\right\} \\
\geq \max\Bigg\{\frac{8}{3}\sqrt{2(1+\langle\phi_1,\phi_K\rangle)} \cdot \gamma_{gap} \cdot \left|\left\langle\psi_1, \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\right\rangle\right|, \\
\frac{8}{3}\gamma_{gap} \cdot \max_{k\notin\{1,K\}} |\langle\psi_k, \phi_k\rangle|, \\
58\gamma_{dyn} \cdot \mu(\Phi,\Psi) \cdot \log K, \\
\frac{50}{3}\gamma_{dyn} \cdot \sqrt{\frac{S\|\Phi\|_{2,2}^2 \log K}{K-S}}\Bigg\}
\end{aligned}$$

and additionally

$$\left|\left\langle \psi_1, \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|}\right\rangle\right| \geq \max\Bigg\{\left|\left\langle \psi_1, \frac{\phi_1-\phi_K}{\|\phi_1-\phi_K\|}\right\rangle\right|, \\ \frac{2}{\sqrt{2(1+\langle\phi_1,\phi_K\rangle)}\cdot(1-\gamma_{gap})-\frac{20}{16}}\cdot\gamma_{gap}\cdot\max_{k\notin\{1,K\}}|\langle\psi_k,\phi_k\rangle| \\ -\frac{\sqrt{2(1+\langle\phi_1,\phi_K\rangle)}}{2}c(S)\left|\left\langle \psi_1, \frac{\phi_1-\phi_K}{\|\phi_1-\phi_K\|}\right\rangle\right|\Bigg\},$$

then

$$\mathbb{P}(\mathcal{R}) \leq \frac{16(S+1)+S^3}{2K^3} \quad \textit{and} \quad \mathbb{P}(\mathcal{R}_{1K}) \leq \frac{S^2}{K^2}\cdot\left(\frac{2}{K^2}+\frac{12}{KS}+\frac{1}{2}\right),$$

where for all $k \in \mathbb{K}\setminus\{1,K\}$

$$\begin{aligned}\mathcal{R} &:= \Big\{(I,\sigma,c) : \Big[\Big(\hat{I}\neq I \vee \operatorname{sign}(\langle\psi_k,y\rangle)\neq\sigma(k)\Big)\wedge k\in I\Big] \vee \Big[k\in\hat{I}\wedge k\notin I\Big]\Big\},\\ \mathcal{R}_{1K} &:= \Big\{(I,\sigma,c) : \Big[\Big(\hat{I}\neq I \vee \operatorname{sign}(\langle\psi_1,y\rangle)\neq\sigma(1)\Big)\wedge(1\in I\wedge K\notin I)\Big]\\ &\qquad \vee \Big[\Big(\hat{I}\neq I_K \vee \operatorname{sign}(\langle\psi_1,y\rangle)\neq\sigma(K)\Big)\wedge(1\notin I\wedge K\in I)\Big]\\ &\qquad \vee \Big[\Big(\hat{I}\neq I_{1K} \vee \operatorname{sign}(\langle\psi_1,y\rangle)\neq\sigma(1)\vee\operatorname{sign}(\langle\psi_1,y\rangle)\neq\sigma(K)\Big)\wedge(1,K\in I)\Big]\\ &\qquad \vee \Big[1\in\hat{I}\wedge 1,K\notin I\Big]\Big\}.\end{aligned} \tag{16}$$

Proof

To prove the statement, we will observe the cases $k = 1$ and $k \in \mathbb{K}\setminus\{1,K\}$ separately. Both cases have the same approach, but in case $k = 1$ calculations slightly simplify. Throughout the proof, we will use the following abbreviations:

$$B := \|\Phi\|_{2,2}^2 \quad \text{and} \quad \hat{\mu} := \max\left\{\max_{1\neq i\neq j}|\langle\psi_i,\phi_j\rangle|, \max_{1\neq j\neq K}|\langle\psi_1,\phi_j\rangle|\right\}.$$

We will show with high probability for a signal y following the model in 2 with $k \in I$ that thresholding recovers the correct support I and the correct sign, meaning $\operatorname{sign}(\langle\psi_k,y\rangle) = \sigma(k)$.

Step 1: Failure Probability of the recovery of I or the correct sign $\sigma(k)$ for $k \in \mathbb{K}\setminus\{1,K\}$: Thresholding has different restrictions for the recovery of the support $\hat{I}$ and sign $\hat{\sigma}(k)$ depending on 1 or K being in the support I and whether $\sigma(1)$ equals $\sigma(K)$. Thus we define the following six sets:

$$\begin{aligned}\mathcal{A}_{1K+} &:= \{(I,\sigma,c) : 1,K\in I\wedge\sigma(1)=\sigma(K)\},\\ \mathcal{A}_{1K-} &:= \{(I,\sigma,c) : 1,K\in I\wedge\sigma(1)\neq\sigma(K)\},\\ \mathcal{A}_{1\vee K+} &:= \{(I,\sigma,c) : [(1\in I,K\notin I)\vee(1\notin I,K\in I)]\wedge\sigma(1)=\sigma(K)\},\\ \mathcal{A}_{1\vee K-} &:= \{(I,\sigma,c) : [(1\in I,K\notin I)\vee(1\notin I,K\in I)]\wedge\sigma(1)\neq\sigma(K)\},\\ \mathcal{A}_{\neg1K+} &:= \{(I,\sigma,c) : 1,K\notin I\wedge\sigma(1)=\sigma(K)\},\\ \mathcal{A}_{\neg1K-} &:= \{(I,\sigma,c) : 1,K\notin I\wedge\sigma(1)\neq\sigma(K)\}.\end{aligned} \tag{17}$$

We estimate the probability that thresholding fails to recover I or the correct sign $\sigma(k)$ by splitting it into the cases above, yielding

$$\begin{aligned}
&\mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle \psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right\}\right) \\
&\quad\le \mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle \psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right\} \cap \mathcal{A}_{1K+}\right) \\
&\qquad + \mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle \psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right\} \cap \mathcal{A}_{1K-}\right) \\
&\qquad + \mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle \psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right\} \cap \mathcal{A}_{1\vee K+}\right) \\
&\qquad + \mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle \psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right\} \cap \mathcal{A}_{1\vee K-}\right) \\
&\qquad + \mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle \psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right\} \cap \mathcal{A}_{\neg 1K+}\right) \\
&\qquad + \mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle \psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right\} \cap \mathcal{A}_{\neg 1K-}\right).
\end{aligned} \tag{18}$$

To ensure the correct recovery of I, we need to have

$$\min_{i \in I \setminus \{K\}} |\langle \psi_i, y\rangle| > \max_{i \notin I \setminus \{K\}} |\langle \psi_i, y\rangle| \quad \text{and for } K \in I \quad |\langle \psi_1, y\rangle| > \max_{i \notin I \setminus \{1\}} |\langle \psi_i, y\rangle|.$$

We recall that $\psi_1 = \alpha_1 \cdot \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|} + \beta_1 \cdot \frac{\phi_1-\phi_K}{\|\phi_1-\phi_K\|} + \omega_1 \cdot z_1$ and expand the inner product of a rescaled signal y with an atom ψ_i for $i \in \mathbb{K} \setminus \{1, K\}$, which yields

$$\begin{aligned}
|\langle \psi_i, \Phi x_{c,p,\sigma}\rangle| &= |\sum_j \sigma(j) c(p(j)) \langle \psi_i, \phi_j\rangle | \\
&= |c(p(i)) \langle \psi_i, \phi_i\rangle + \sigma(i) \sum_{j \neq i} \sigma(j) c(p(j)) \langle \psi_i, \phi_j\rangle | \quad \text{and} \\
|\langle \psi_1, \Phi x_{c,p,\sigma}\rangle| &= |c(p(1)) \langle \psi_1, \phi_1\rangle + \sigma(1)\sigma(K) c(p(K)) \langle \psi_1, \phi_K\rangle + \sigma(1) \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_i, \phi_j\rangle | \\
&= |c(p(1)) \left(\alpha_1 \cdot \frac{1 + \langle \phi_1, \phi_K\rangle}{\|\phi_1 + \phi_K\|} + \beta_1 \frac{1 - \langle \phi_1, \phi_K\rangle}{\|\phi_1 - \phi_K\|}\right) \\
&\quad + \sigma(1)\sigma(K) c(p(K)) \left(\alpha_1 \cdot \frac{1 + \langle \phi_1, \phi_K\rangle}{\|\phi_1 + \phi_K\|} + \beta_1 \frac{-1 + \langle \phi_1, \phi_K\rangle}{\|\phi_1 - \phi_K\|}\right) + \sigma(1) \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_i, \phi_j\rangle | \\
&= |[c(p(1)) + \sigma(1)\sigma(K) c(p(K))] \cdot \alpha_1 \frac{1 + \langle \phi_1, \phi_K\rangle}{\|\phi_1 + \phi_K\|} \\
&\quad [c(p(1)) - \sigma(1)\sigma(K) c(p(K))] \cdot \beta_1 \frac{1 - \langle \phi_1, \phi_K\rangle}{\|\phi_1 - \phi_K\|} + \sigma(1) \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_i, \phi_j\rangle | \\
&= |[c(p(1)) + \sigma(1)\sigma(K) c(p(K))] \cdot \alpha_1 \frac{\sqrt{2(1 + \langle \phi_1, \phi_K\rangle)}}{2} \\
&\quad [c(p(1)) - \sigma(1)\sigma(K) c(p(K))] \cdot \beta_1 \frac{\sqrt{2(1 - \langle \phi_1, \phi_K\rangle)}}{2} + \sigma(1) \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_i, \phi_j\rangle |.
\end{aligned}$$

Depending on whether the index $i \in \mathbb{K} \setminus \{1, K\}$ is in the support I or not, we obtain the following bounds from below resp. above:

$$
\begin{aligned}
i \in I : |\langle \psi_i, \Phi x_{c,p,\sigma} \rangle| &\geq c(S)\alpha_{\min} - \left| \sum_{j \neq i} \sigma(j) c(p(j)) \langle \psi_i, \phi_j \rangle \right|, \\
i \notin I : |\langle \psi_i, \Phi x_{c,p,\sigma} \rangle| &\leq c(S+1)\alpha_{\max} + \left| \sum_{j \neq i} \sigma(j) c(p(j)) \langle \psi_i, \phi_j \rangle \right|,
\end{aligned} \tag{19}
$$

For $i \in \{1, K\}$, we use the sets we defined in (17). We recall that $\alpha_1 \geq \beta_1$ and define $n_{1+K} := \|\phi_1 + \phi_K\| = \sqrt{2(1 + \langle \phi_1, \phi_K \rangle)}$ and $n_{1-K} := \|\phi_1 - \phi_K\| = \sqrt{2(1 - \langle \phi_1, \phi_K \rangle)}$ as well as $\mathbb{I} := (I, \sigma, c)$.

Without loss of generality we assume $\langle \phi_1, \phi_K \rangle \geq 0$, which implies $n_{1+K} \geq n_{1-K}$ and get

$\mathcal{I} \in \mathcal{A}_{1K+}$:

$$\begin{aligned} |\langle \psi_1, \Phi x_{c,p,\sigma} \rangle| &\geq [c(S-1)+c(S)]\alpha_1 \frac{n_{1+K}}{2} \\ &\quad + [c(S-1)-c(S)]\beta_1 \frac{n_{1-K}}{2} - \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right| \\ &\geq c(S)\alpha_{\min} n_{1+K} - \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right|, \end{aligned}$$

$\mathcal{I} \in \mathcal{A}_{1\vee K+}$:

$$\begin{aligned} |\langle \psi_1, \Phi x_{c,p,\sigma} \rangle| &\geq [c(S)+c(K)]\alpha_1 \frac{n_{1+K}}{2} \\ &\quad + [c(S)-c(K)]\beta_1 \frac{n_{1-K}}{2} - \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right| \\ &\geq c(S)(\alpha_{\min} \frac{n_{1+K}}{2} + \beta_1 \frac{n_{1-K}}{2}) - \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right|, \end{aligned}$$

$\mathcal{I} \in \mathcal{A}_{1\vee K-}$:

$$\begin{aligned} |\langle \psi_1, \Phi x_{c,p,\sigma} \rangle| &\geq [c(S)-c(S+1)]\alpha_1 \frac{n_{1+K}}{2} \\ &\quad + [c(S)+c(S+1)]\beta_1 \frac{n_{1-K}}{2} - \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right|, \end{aligned}$$

$\mathcal{I} \in \mathcal{A}_{\neg 1K+}$:

$$\begin{aligned} |\langle \psi_1, \Phi x_{c,p,\sigma} \rangle| &\leq [c(S+1)+c(S+2)]\alpha_1 \frac{n_{1+K}}{2} \\ &\quad + [c(S+1)-c(S+2)]\beta_1 \frac{n_{1-K}}{2} + \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right| \\ &\leq c(S+1)\alpha_1 n_{1+K} + [c(S+1)-c(S+2)]\beta_1 \frac{n_{1-K}}{2} + \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right|, \end{aligned}$$

$\mathcal{I} \in \mathcal{A}_{\neg 1K-}$:

$$\begin{aligned} |\langle \psi_1, \Phi x_{c,p,\sigma} \rangle| &\leq [c(S+1)-c(K)]\alpha_1 \frac{n_{1+K}}{2} \\ &\quad + [c(S+1)+c(K)]\beta_1 \frac{n_{1-K}}{2} + \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right| \\ &\leq c(S+1)\left(\alpha_1 \frac{n_{1+K}}{2} + \beta_1 \frac{n_{1-K}}{2}\right) + \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right|. \end{aligned} \tag{20}$$

We point out that the worst case $I \in \mathcal{A}_{1K-}$ does not need to be covered above, since thresholding can never recover the correct signs $\sigma(1)$ and $\sigma(K)$ together. We achieve recovery of I, if

$$\max_{i\in\mathbb{K}\setminus\{1,K\}} \left\{ \left| \sum_{j\notin\{1,K\}} \sigma(j)c(p(j)) \langle \psi_1, \phi_j \rangle \right|, \left| \sum_{j\neq i} \sigma(j)c(p(j)) \langle \psi_i, \phi_j \rangle \right| \right\} =$$

$$\max_{i\in\mathbb{K}\setminus\{K\}} \left| \sum_{j\neq i} \sigma(j)c(p(j)) \langle \psi_i, \phi_j \rangle - \sigma(K)c(p(K)) \langle \psi_1, \phi_K \rangle \mathbb{1}_{(i=1)\wedge(j=K)} \right| < \theta \cdot c(S)\alpha_{\min}, \tag{21}$$

where θ ensures the following conditions depending on the cases in (17):

$$\begin{aligned}
&\mathcal{I} \in \mathcal{A}_{1K+}: \\
&\quad c(S)\alpha_{\min} n_{1+K} - \theta c(S)\alpha_{\min} \ \geq \ c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \ \overset{!}{\geq} \ c(S+1)\alpha_{\max} + \theta c(S)\alpha_{\min}, \\
&\mathcal{I} \in \mathcal{A}_{1\vee K+}: \\
&\quad \psi_1 : c(S)\left(\alpha_1 \frac{n_{1+K}}{2} + \beta_1 \frac{n_{1-K}}{2}\right) - \theta c(S)\alpha_{\min} \overset{!}{\geq} c(S+1)\alpha_{\max} + \theta c(S)\alpha_{\min}, \\
&\quad \Rightarrow c(S)\left(\alpha_1 \frac{n_{1+K}}{2} + \beta_1 \frac{n_{1-K}}{2}\right) - \theta c(S)\alpha_1 \overset{!}{\geq} c(S+1)\alpha_{\max} + \theta c(S)\alpha_1, \\
&\quad \psi_i : c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \overset{!}{\geq} c(S+1)\alpha_{\max} + \theta c(S)\alpha_{\min}, \\
&\mathcal{I} \in \mathcal{A}_{1\vee K-}: \\
&\quad \psi_1 : \frac{n_{1+K}}{2}(c(S) - c(S+1))\alpha_1 + \frac{n_{1-K}}{2}(c(S) + c(S+1))\beta_1 - \theta c(S)\alpha_{\min} \overset{!}{\geq} \\
&\qquad\qquad c(S+1)\alpha_{\max} + \theta c(S)\alpha_{\min}, \\
&\quad \Rightarrow \frac{n_{1+K}}{2}(c(S) - c(S+1))\alpha_1 + \frac{n_{1-K}}{2}(c(S) + c(S+1))\beta_1 - \theta c(S)\alpha_1 \overset{!}{\geq} \\
&\qquad\qquad c(S+1)\alpha_{\max} + \theta c(S)\alpha_1, \\
&\quad \psi_i : c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \overset{!}{\geq} c(S+1)\alpha_{\max} + \theta c(S)\alpha_{\min}, \\
&\mathcal{I} \in \mathcal{A}_{\neg 1K+}: \\
&\quad c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \overset{!}{\geq} \\
&\qquad\qquad c(S+1)\max\left\{\alpha_{\max}, \alpha_1 n_{1+K} + \frac{c(S+1) - c(S+2)}{c(S+1)}\beta_1 \frac{n_{1-K}}{2}\right\} + \theta c(S)\alpha_{\min}, \\
&\mathcal{I} \in \mathcal{A}_{\neg 1K-}: \\
&\quad c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \overset{!}{\geq} c(S+1)\max\left\{\alpha_1 \frac{n_{1+K}}{2} + \beta_1 \frac{n_{1-K}}{2}, \alpha_{\max}\right\} + \theta c(S)\alpha_{\min}.
\end{aligned} \tag{22}$$

We can simplify these conditions to

$$\begin{aligned}
\alpha_1 :&\quad \frac{n_{1+K}}{2}(c(S) - c(S+1))\alpha_1 + \frac{n_{1-K}}{2}(c(S) + c(S+1))\beta_1 - \theta c(S)\alpha_1 \overset{!}{\geq} c(S+1)\alpha_{\max} + \theta c(S)\alpha_1, \\
\alpha_{\min} :&\quad c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \overset{!}{\geq} \\
&\qquad c(S+1)\max\left\{\alpha_{\max}, \alpha_1 n_{1+K} + \frac{c(S+1) - c(S+2)}{c(S+1)}\beta_1 \frac{n_{1-K}}{2}\right\} + \theta c(S)\alpha_{\min}.
\end{aligned} \tag{23}$$

The conditions above also guarantees the recovery of the correct sign $\sigma(i)$ for all $i \in I$, so in particular the recovery of $\sigma(k)$. We define $c_{1K} := \sigma(K)c(p(K)) \langle \psi_1, \phi_K \rangle \mathbb{1}_{(i=1)\wedge(j=K)}$ and establish the following estimate for $k \in \mathbb{K} \setminus \{1, K\}$:

$$\mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle \psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right\} \cap \mathcal{A}_{1K+}\right)$$
$$\leq \sum_{i \in \mathbb{K}\setminus\{K\}} \mathbb{P}\left(\left\{(I,\sigma,c) : \left|\sum_{j\neq i} \sigma(j)c(p(j))\langle \psi_i, \phi_j\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \wedge k \in I\right\} \cap \mathcal{A}_{1K+}\right).$$

For simplicity we will write for the further proof $\mathbb{P}\left(\left\{(I,\sigma,c) : \left|\sum_{j\neq i} \sigma(j)c(p(j))\langle \psi_i, \phi_j\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \wedge k \in I\right\} \cap \mathcal{A}_{1K+}\right)$ as $\mathbb{P}\left(\left|\sum_{j\neq i} \sigma(j)c(p(j))\langle \psi_i, \phi_j\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min}, k \in I, \mathcal{A}_{1K+}\right)$. Before we address the other cases in (17) we find an upper bound of the probability above. Thus we rewrite the probabilities as conditional probabilities, yielding

$$\mathbb{P}\left(\left|\sum_{j\neq i} \sigma(j)c(p(j))\langle \psi_i, \phi_j\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min}, k \in I, \mathcal{A}_{1K+}\right)$$
$$\leq \sum_{(\ell_k,\ell_1,\ell_K)\in\mathbb{S}^3} \mathbb{P}\left(\left|\sum_{j\neq i} \sigma(j)c(p(j))\langle \psi_i, \phi_j\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right) \times$$
$$\times \mathbb{P}((p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)) \tag{24}$$

Next we split the sum into two parts; one over $j \in I\setminus\{i,k,1,K\}$, that captures most of the energy, and the other over $j \in (I^c \cup \{k,1,K\})\setminus\{i\}$. For $m_1 \in (0,1)$ and $m_2 = 1 - m_1$ we have

$$\mathbb{P}\left(\left|\sum_{j\neq i} \sigma(j)c(p(j))\langle \psi_i, \phi_j\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right)$$
$$\leq \mathbb{P}\left(\left|\sum_{j\in I\setminus\{i,k,1,K\}} \sigma(j)c(p(j))\langle \psi_i, \phi_j\rangle - c_{1K}\right| \geq m_1\theta c(S)\alpha_{\min}\right.$$
$$\left.\mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right)$$
$$+ \mathbb{P}\left(\left|\sum_{j\in (I^c\cup\{k,1,K\})\setminus\{i\}} \sigma(j)c(p(j))\langle \psi_i, \phi_j\rangle - c_{1K}\right| \geq m_2\theta c(S)\alpha_{\min}\right.$$
$$\left.\mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right). \tag{25}$$

The first term we estimate using Lemma 4 and the second term using Hoeffding's inequality, [14]. With some small simplifications we get for all i, including k,

$$\mathbb{P}\left(\left|\sum_{j\neq i}\sigma(j)c(p(j))\left\langle\psi_i,\phi_j\right\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right)$$

$$\leq 2\exp\left(\frac{-\left(m_1\theta c(S)\alpha_{\min}\right)^2}{2\left(c(1)\hat{\mu}\cdot m_1\theta c(S)\alpha_{\min} + \|c(\mathbb{S})\|_2^2\frac{B}{K-S}\right)}\right) + 2\exp\left(\frac{-\left(m_2\theta c(S)\alpha_{\min}\right)^2}{2\hat{\mu}^2\left(c(\ell_k)^2+c(\ell_1)^2+c(\ell_K)^2+\|c\left(\mathbb{S}^c\right)\|_2^2\right)}\right)$$

$$\leq 2\exp\left(-\frac{1}{4}\min\left\{\frac{c(S)m_1\theta\alpha_{\min}}{c(1)\hat{\mu}}, \frac{(K-S)\left(m_1\theta c(S)\alpha_{\min}\right)^2}{B\|c(\mathbb{S})\|_2^2}\right\}\right)$$

$$+ 2\exp\left(-\frac{1}{4}\min\left\{\frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{3c(1)^2\hat{\mu}^2}, \frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{\hat{\mu}^2\left\|c\left(\mathbb{S}^c\right)\right\|_2^2}\right\}\right)$$

We use the fact that every permutation is equally likely, yielding $\mathbb{P}((p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1) = \sigma(K)) = \frac{1}{2K(K-1)(K-2)}$ and substitute the expression above into (24):

$$\mathbb{P}\left(\left(\hat{I}\neq I \vee \operatorname{sign}\left(\left\langle\psi_k,y\right\rangle\right)\neq\sigma(k)\right), k\in I, \mathcal{A}_{1K+}\right)$$

$$\leq \frac{(K-1)S(S-1)(S-2)}{K(K-1)(K-2)}\exp\left(-\frac{1}{4}\min\left\{\frac{c(S)m_1\theta\alpha_{\min}}{c(1)\hat{\mu}}, \frac{(K-S)\left(m_1\theta c(S)\alpha_{\min}\right)^2}{B\|c(\mathbb{S})\|_2^2}\right\}\right)$$

$$+ \frac{(K-1)S(S-1)(S-2)}{K(K-1)(K-2)}\exp\left(-\frac{1}{4}\min\left\{\frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{3c(1)^2\hat{\mu}^2}, \frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{\hat{\mu}^2\left\|c\left(\mathbb{S}^c\right)\right\|_2^2}\right\}\right),$$

where θ has to ensure (23) and $m_1+m_2=1$. From this, whenever

$$\alpha_{\min} \geq \max\left\{\frac{1}{1-2\theta}\frac{c(S+1)}{c(S)}\max\left\{\alpha_{\max}, \alpha_1 n_{1+K} + \frac{c(S+1)-c(S+2)}{c(S+1)}\beta_1\frac{n_{1-K}}{2}\right\},\right.$$
$$\frac{4n}{m_1\theta_1}\frac{c(1)}{c(S)}\hat{\mu}\log K, \frac{2\sqrt{n}}{m_1\theta_1}\frac{\|c(\mathbb{S})\|_2}{c(S)}\sqrt{\frac{B\log K}{K-S}}\frac{2\sqrt{n}}{(1-m_1)\theta_1}\frac{3c(1)}{c(S)}\hat{\mu}\sqrt{\log K},$$
$$\left.\frac{2\sqrt{n}}{(1-m_1)\theta_1}\frac{\left\|c\left(\mathbb{S}^c\right)\right\|_2}{c(S)}\hat{\mu}\sqrt{\log K}\right\} \quad \text{and}$$
$$\alpha_1 \geq \frac{2}{n_{1+K}\frac{c(S)-c(S+1)}{c(S)} - 4\theta}\frac{c(S+1)}{c(S)}\left(\alpha_{\max} - \frac{c(S)+c(S+1)}{c(S+1)}\frac{n_{1-K}}{2}\beta_1\right). \tag{26}$$

we get that

$$\mathbb{P}\left(\left(\hat{I}\neq I \vee \operatorname{sign}\left(\left\langle\psi_k,y\right\rangle\right)\neq\sigma(k)\right), k\in I, \mathcal{A}_{1K+}\right) \leq 2\frac{(K-1)S(S-1)(S-2)}{K(K-1)(K-2)}\cdot K^{-n}.$$

For the other probabilities in (18) we follow the same calculations, except for some minor changes. We outline the difference for the case $I\in\mathcal{A}_{1\vee K+}$ to see the pattern. Without loss of generality we say $1\in I$ and $K\notin I$ and obtain $(\ell_k,\ell_1,\ell_K)\in\mathbb{S}^2\times(\mathbb{K}\setminus\mathbb{S})$ in equation (24), meaning

$$\mathbb{P}\left(\left(\hat{I}\neq I \vee \operatorname{sign}\left(\left\langle\psi_k,y\right\rangle\right)\neq\sigma(k)\right), k\in I, \mathcal{A}_{1\vee K+}\right)$$
$$\leq \sum_{(\ell_k,\ell_1,\ell_K)\in\mathbb{S}^2\times(\mathbb{K}\setminus\mathbb{S})}\mathbb{P}\left(\left|\sum_{j\neq i}\sigma(j)c(p(j))\left\langle\psi_i,\phi_j\right\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min}\right.$$
$$\left.\mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right)\times$$
$$\times\,\mathbb{P}((p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)).$$

We follow the exact same steps as in (25) by splitting the sum into the part $j \in I \setminus \{i,k,1\}$ and the other over $j \in (I^c \cup \{k,1\}) \setminus \{i\}$ before applying Lemma 4 and Hoeffding's inequality, [14], yielding

$$\begin{aligned}
&\mathbb{P}\left(\left|\sum_{j\neq i}\sigma(j)c(p(j))\langle\psi_i,\phi_j\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right)\\
&\quad\leq \mathbb{P}\left(\left|\sum_{j\in I\setminus\{i,k,1\}}\sigma(j)c(p(j))\langle\psi_i,\phi_j\rangle - c_{1K}\right| \geq m_1\theta c(S)\alpha_{\min}\right.\\
&\qquad\qquad\qquad\qquad\qquad\qquad\left.\mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right)\\
&\qquad+\mathbb{P}\left(\left|\sum_{j\in (I^c\cup\{k,1\})\setminus\{i\}}\sigma(j)c(p(j))\langle\psi_i,\phi_j\rangle - c_{1K}\right| \geq m_2\theta c(S)\alpha_{\min}\right.\\
&\qquad\qquad\qquad\qquad\qquad\qquad\left.\mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right)
\end{aligned}$$

$$\begin{aligned}
&\mathbb{P}\left(\left|\sum_{j\neq i}\sigma(j)c(p(j))\langle\psi_i,\phi_j\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right)\\
&\quad\leq 2\exp\left(\frac{-\left(m_1\theta c(S)\alpha_{\min}\right)^2}{2\left(c(1)\hat{\mu}\cdot m_1\theta c(S)\alpha_{\min} + \|c(\mathbb{S})\|_2^2\frac{B}{K-S}\right)}\right) + 2\exp\left(\frac{-\left(m_2\theta c(S)\alpha_{\min}\right)^2}{2\hat{\mu}^2\left(c(\ell_k)^2 + c(\ell_1)^2 + \|c\left(\mathbb{S}^c\right)\|_2^2\right)}\right)\\
&\quad\leq 2\exp\left(-\frac{1}{4}\min\left\{\frac{c(S)m_1\theta\alpha_{\min}}{c(1)\hat{\mu}}, \frac{(K-S)\left(m_1\theta c(S)\alpha_{\min}\right)^2}{B\|c(\mathbb{S})\|_2^2}\right\}\right)\\
&\qquad+2\exp\left(-\frac{1}{4}\min\left\{\frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{2c(1)^2\hat{\mu}^2}, \frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{\hat{\mu}^2\|c\left(\mathbb{S}^c\right)\|_2^2}\right\}\right).
\end{aligned}$$

We can see that with the same conditions as in (26) we obtain

$$\mathbb{P}\left(\left(\hat{I}\neq I \vee \operatorname{sign}\left(\langle\psi_k,y\rangle\right)\neq\sigma(k)\right), k\in I, \mathcal{A}_{1\vee K+}\right) \leq 4\frac{(K-1)S(S-1)(K-S)}{K(K-1)(K-2)}\cdot K^{-n}.$$

In the same manner we proceed with the cases $\mathcal{I}\in\mathcal{A}_{1\vee K-}, I\in\mathcal{A}_{\neg 1K+}$ and $\mathcal{I}\in\mathcal{A}_{\neg 1K-}$. The most problematic case $\mathcal{I}\in\mathcal{A}_{1K-}$, where thresholding cannot recover the correct support and sign, we crudely estimate by

$$\begin{aligned}
&\mathbb{P}\left(\left(\hat{I}\neq I \vee \operatorname{sign}\left(\langle\psi_k,y\rangle\right)\neq\sigma(k)\right), k\in I, \mathcal{A}_{1K-}\right)\\
&\quad\leq \mathbb{P}\left(\mathcal{A}_{1K-}\right) \leq \binom{K-3}{S-3}\Big/\left(2\cdot\binom{K}{S}\right) = \frac{S(S-1)(S-2)}{2K(K-1)(K-2)}.
\end{aligned}\tag{27}$$

Putting all six cases from (18) together, for any $k\in\mathbb{K}\setminus\{1,K\}$ we get

$$\mathbb{P}\left(\left(\hat{I}\neq I \vee \operatorname{sign}\left(\langle\psi_k,y\rangle\right)\neq\sigma(k)\right), k\in I\right) \leq 2S\left(\frac{S^2}{K^2} + 4\frac{S(K-S)}{K^2} + 2\frac{(K-S)^2}{K^2}\right)\cdot K^{-n} + \frac{S^3}{2K^3}.\tag{28}$$

Step 2: Probability of wrongly recovering k for $k\notin I\cup\{1,K\}$ – $\mathbb{P}\left(k\in\hat{I}\mid k\notin I\right)$:
As a second step we will bound the probability of wrongly recovering an atom ψ_k when it is not in the generating support, meaning $k\notin I$. With the sets of (17), we split the probability of thresholding

recovering k when it is not in the generating support I, yielding

$$\begin{aligned}\mathbb{P}\left(\left\{(I,\sigma,c) : k \in \hat{I} \wedge k \notin I\right\}\right) \leq \mathbb{P}&\left(\left\{(I,\sigma,c) : k \in \hat{I} \wedge k \notin I\right\} \cap \mathcal{A}_{1K+}\right)\\
&+\mathbb{P}\left(\left\{(I,\sigma,c) : k \in \hat{I} \wedge k \notin I\right\} \cap \mathcal{A}_{1K-}\right)\\
&+\mathbb{P}\left(\left\{(I,\sigma,c) : k \in \hat{I} \wedge k \notin I\right\} \cap \mathcal{A}_{1\vee K+}\right)\\
&+\mathbb{P}\left(\left\{(I,\sigma,c) : k \in \hat{I} \wedge k \notin I\right\} \cap \mathcal{A}_{1\vee K-}\right)\\
&+\mathbb{P}\left(\left\{(I,\sigma,c) : k \in \hat{I} \wedge k \notin I\right\} \cap (\mathcal{A}_{\neg 1K+} \cup \mathcal{A}_{\neg 1K-})\right)\end{aligned}$$

A sufficient condition for not recovering $k \in \mathbb{K} \setminus \{1, K\}$ is that

$$\min_{i \in I\setminus\{K\}} |\langle\psi_i, y\rangle| > |\langle\psi_k, y\rangle| \quad \text{and for } K \in I \quad |\langle\psi_1, y\rangle| > |\langle\psi_k, y\rangle|.$$

Again we expand this condition and observe each case as in (19) and (20), yielding for $k, i \in \mathbb{K}\setminus\{1, K\}$

$$\begin{aligned}&k \notin I : |\langle\psi_k, \Phi x_{c,p,\sigma} + r\rangle| \leq c(S+1)\alpha_k - \left|\sum_{j\neq k} \sigma(j)c(p(j))\langle\psi_k, \phi_j\rangle\right|,\\
&i \in I : |\langle\psi_i, \Phi x_{c,p,\sigma} + r\rangle| \geq c(S)\alpha_{\min} - \left|\sum_{j\neq i} \sigma(j)c(p(j))\langle\psi_i, \phi_j\rangle\right|,\\
&\mathcal{I} \in \mathcal{A}_{1K+} :\\
&\qquad |\langle\psi_1, \Phi x_{c,p,\sigma} + r\rangle| \geq c(S)\alpha_{\min} n_{1+K} - \left|\sum_{j\notin\{1,K\}} \sigma(j)c(p(j))\langle\psi_1, \phi_j\rangle\right|,\\
&\mathcal{I} \in \mathcal{A}_{1\vee K+} :\\
&\qquad |\langle\psi_1, \Phi x_{c,p,\sigma} + r\rangle| \geq c(S)(\alpha_{\min}\frac{n_{1+K}}{2} + \beta_1\frac{n_{1-K}}{2}) - \left|\sum_{j\notin\{1,K\}} \sigma(j)c(p(j))\langle\psi_1, \phi_j\rangle\right|,\\
&\mathcal{I} \in \mathcal{A}_{1\vee K-} :\\
&\qquad |\langle\psi_1, \Phi x_{c,p,\sigma} + r\rangle| \geq [c(S) - c(S+1)]\alpha_1\frac{n_{1+K}}{2}\\
&\qquad\qquad + [c(S) + c(S+1)]\beta_1\frac{n_{1-K}}{2} - \left|\sum_{j\notin\{1,K\}} \sigma(j)c(p(j))\langle\psi_1, \phi_j\rangle\right|,\end{aligned}$$

We bound the probability of thresholding recovering k when it is not in the generating support I similar as in step 1. Thus we get for all $i \in \mathbb{K}$

$$\max_{i\in\mathbb{K}\setminus\{K\}} \left|\sum_{j\neq i} \sigma(j)c(p(j))\langle\psi_i, \phi_j\rangle - c_{1K}\right| < \theta \cdot c(S)\alpha_{\min},$$

where θ ensures slightly weaker conditions as in (23), namely

$$\begin{aligned}&\alpha_1 : \quad \frac{n_{1+K}}{2}(c(S) - c(S+1))\alpha_1 + \frac{n_{1-K}}{2}(c(S) + c(S+1))\beta_1 - \theta c(S)\alpha_1 \overset{!}{\geq} c(S+1)\alpha_k + \theta c(S)\alpha_1,\\
&\alpha_{\min} : \quad c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \overset{!}{\geq} c(S+1)\alpha_k, +\theta c(S)\alpha_{\min}.\end{aligned}$$

Starting with the case $\mathcal{I} \in \mathcal{A}_{1K+}$, we bound the probability of thresholding recovering k when it is not in the generating support I as

$$
\begin{aligned}
&\mathbb{P}\left(k \in \hat{I}, k \notin I, \mathcal{A}_{1K+}\right) \\
&\quad\leq \sum_{(\ell_k,\ell_1,\ell_K)\in(\mathbb{K}\setminus\mathbb{S})\times\mathbb{S}^2} \mathbb{P}\left(k \in \hat{I} \mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right) \\
&\quad\leq \frac{1}{2K(K-1)(K-2)} \times \\
&\qquad\times \sum_{(\ell_k,\ell_1,\ell_K)\in(\mathbb{K}\setminus\mathbb{S})\times\mathbb{S}^2} \sum_{i\in I\setminus\{K\}\cup\{1,k\}} \mathbb{P}\left(\left|\sum_{j\neq i} \sigma(j)c(p(j))\langle\psi_i,\phi_j\rangle - c_{1K}\right| \geq \theta\cdot c(S)\alpha_{\min}\right. \\
&\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\left. \mid (p(k),p(1),p(K)) = (\ell_k,\ell_1,\ell_K), \sigma(1)=\sigma(K)\right).
\end{aligned}
$$

Using the same splitting technique of the sum as in step 1, namely one part over $j \in I\setminus\{i,1,K\}$ and the other over $j \in (I^c \cup \{1,K\})\setminus\{i\}$, and applying Lemma 4 and Hoeffding's inequality,[14], we get

$$
\begin{aligned}
&\mathbb{P}\left(k \in \hat{I}, k \notin I, \mathcal{A}_{1K+}\right) \\
&\quad\leq 2\frac{(K-S)S(S-1)(S+1)}{2K(K-1)(K-2)} \exp\left(-\frac{1}{4}\min\left\{\frac{c(S)m_1\theta\alpha_{\min}}{c(1)\hat{\mu}}, \frac{(K-S)\left(m_1\theta c(S)\alpha_{\min}\right)^2}{B\|c(\mathbb{S})\|_2^2}\right\}\right) \\
&\quad + 2\frac{(K-S)S(S-1)(S+1)}{2K(K-1)(K-2)} \exp\left(-\frac{1}{4}\min\left\{\frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{2c(1)^2\hat{\mu}^2}, \frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{\hat{\mu}^2\left\|c\left(\mathbb{S}^c\right)\right\|_2^2}\right\}\right).
\end{aligned}
$$

In order to have this probability sufficiently small, we require

$$
\begin{aligned}
\alpha_{\min} \geq \max&\left\{\frac{1}{1-2\theta}\frac{c(S+1)}{c(S)}\alpha_k, \frac{4n}{m_1\theta_1}\frac{c(1)}{c(S)}\hat{\mu}\log K,\right. \\
&\left.\frac{2\sqrt{n}}{m_1\theta_1}\frac{\|c(\mathbb{S})\|_2}{c(S)}\sqrt{\frac{B\log K}{K-S}}\frac{2\sqrt{n}}{(1-m_1)\theta_1}\frac{2c(1)}{c(S)}\hat{\mu}\sqrt{\log K}, \frac{2\sqrt{n}}{(1-m_1)\theta_1}\frac{\left\|c\left(\mathbb{S}^c\right)\right\|_2}{c(S)}\hat{\mu}\sqrt{\log K}\right\} \quad \text{and} \\
\alpha_1 \geq& \frac{2}{n_{1+K}\frac{c(S)-c(S+1)}{c(S)} - 4\theta}\frac{c(S+1)}{c(S)}\left(\alpha_k - \frac{c(S)+c(S+1)}{c(S+1)}\frac{n_{1-K}}{2}\beta_1\right).
\end{aligned} \tag{29}
$$

Thus we get

$$
\mathbb{P}\left(k \in \hat{I}, k \notin I, \mathcal{A}_{1K+}\right) \leq 2\frac{(S+1)(K-S)S(S-1)}{K(K-1)(K-2)} \cdot K^{-n}.
$$

As in step 1 we do the same calculations with $\mathcal{I} \in \mathcal{A}_{1\vee K+}, \mathcal{I} \in \mathcal{A}_{1\vee K-}$ and $\mathcal{I} \in \mathcal{A}_{\neg 1K+} \cup \mathcal{A}_{\neg 1K-}$. Since $\mathbb{P}\left(k \in \hat{I}, k \notin I, \mathcal{A}_{1K-}\right) \leq \mathbb{P}\left(\mathcal{A}_{1K-}\right)$ we have already established the estimate in (27). Putting all the bounds together with the bound of $\mathbb{P}\left(\left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle\psi_k, y\rangle\right) \neq \sigma(k)\right), k \in I\right)$ in (28) as well as using that $\frac{S-\ell}{K-\ell} \leq \frac{S}{K}$ for $\ell < S$, we get for $k \in \mathbb{K}\setminus\{1,K\}$

$$
\begin{aligned}
&\mathbb{P}\left(\left\{(I,\sigma,c) : \left[\left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle\psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right] \vee \left(k \in \hat{I} \wedge k \notin I\right)\right\}\right) \\
&\quad\leq \left[S\left(\frac{2S^2}{K^2} + \frac{8S(K-S)}{K^2} + \frac{4(K-S)^2}{K^2}\right)\right. \\
&\qquad\qquad\left. +(S+1)\left(\frac{2S^2(K-S)}{K^3} + \frac{8S(K-S)^2}{K^3} + \frac{4(K-S)^3}{K^3}\right)\right] \cdot K^{-n} + \frac{S^3}{2K^3}.
\end{aligned} \tag{30}
$$

3: Failure Probability of the recovery of I or the correct sign $\sigma(1)$ or $\sigma(K)$ or wrongly recovering 1

In this step we focus on whenever 1 or K are in the support. We recall that thresholding can only recover the correct support I, if $K \notin I$, since Ψ has one less atom than Φ and thus covers the missing atom ϕ_K with the double atom $\psi_1 = \alpha_1 \cdot \frac{\phi_1+\phi_K}{\|\phi_1+\phi_K\|} + \beta_1 \cdot \frac{\phi_1-\phi_K}{\|\phi_1-\phi_K\|} + \omega_1 \cdot z_1$. Hence we say that thresholding also recovers the correct support I with $K \in I$ and $1 \notin I$, if $\hat{I} = I \setminus \{K\} \cup \{1\} =: I_K$. Whenever $1, K \in I$ thresholding cannot recover the correct support, since $|\hat{I}|$ is enforced to have S elements and hence has to choose some $i \in I^c$. Let $I_{1K} \in \{\tilde{I} \in \mathbb{K}^S : \tilde{I} = I \backslash \{K\} \cup \{i\}$ for some $i \in I^c\}$, then we get the best recovery of the support, whenever $\hat{I} = I_{1K}$. So thresholding fails whenever I is in

$$\begin{aligned}\mathcal{R}_{1K} := \Big\{(I,\sigma,c) : &\Big[\Big(\hat{I} \neq I \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(1)\Big) \wedge 1 \in I \wedge K \notin I\Big] \\ &\vee \Big[\Big(\hat{I} \neq I_K \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(K)\Big) \wedge 1 \notin I \wedge K \in I\Big] \\ &\vee \Big[\Big(\hat{I} \neq I_{1K} \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(1) \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(K)\Big) \wedge 1, K \in I\Big] \\ &\qquad \vee \Big[1 \in \hat{I} \wedge 1, K \notin I\Big]\Big\}.\end{aligned}$$

We start by splitting the probability of $\mathbb{P}(\mathcal{R}_{1K})$ into seven cases

$$\begin{aligned}\mathbb{P}(\mathcal{R}_{1K}) \leq\ & \mathbb{P}\Big(\Big\{(I,\sigma,c) : \Big(\hat{I} \neq I \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(1) = \sigma(K)\Big) \wedge 1 \in I \wedge K \notin I\Big\}\Big) \\ &+ \mathbb{P}\Big(\Big\{(I,\sigma,c) : \Big(\hat{I} \neq I \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(1) \neq \sigma(K)\Big) \wedge 1 \in I \wedge K \notin I\Big\}\Big) \\ &+ \mathbb{P}\Big(\Big\{(I,\sigma,c) : \Big(\hat{I} \neq I_K \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(K) = \sigma(1)\Big) \wedge 1 \notin I \wedge K \in I\Big\}\Big) \\ &+ \mathbb{P}\Big(\Big\{(I,\sigma,c) : \Big(\hat{I} \neq I_K \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(K) \neq \sigma(1)\Big) \wedge 1 \notin I \wedge K \in I\Big\}\Big) \\ &+ \mathbb{P}\Big(\Big\{(I,\sigma,c) : \Big(\hat{I} \neq I_{1K} \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(1) = \sigma(K)\Big) \wedge 1, K \in I\Big\}\Big) \qquad (31) \\ &+ \mathbb{P}\Big(\Big\{(I,\sigma,c) : \Big(\hat{I} \neq I_{1K} \vee \operatorname{sign}(\langle\psi_1, y\rangle) \neq \sigma(1) \neq \sigma(K)\Big) \wedge 1, K \in I\Big\}\Big) \\ &+ \mathbb{P}\Big(\Big\{(I,\sigma,c) : 1 \in \hat{I} \wedge 1, K \notin I\Big\}\Big). \qquad (32)\end{aligned}$$

We see that these cases resemble the sets of (18), except for the special supports I_K and I_{1K} and for wether $\operatorname{sign}(\langle\psi_1, y\rangle)$ is compared with $\sigma(1)$ or $\sigma(K)$. Again as a sufficient condition for the correct recovery of I, with $1 \in I$ or $K \in I$ we get

$$\min_{i \in I \backslash \{K\}} |\langle\psi_i, y\rangle| > \max_{i \notin I} |\langle\psi_i, y\rangle| \quad \text{and for } K \in I \quad |\langle\psi_1, y\rangle| > \max_{i \notin I \backslash \{1\}} |\langle\psi_i, y\rangle|.$$

We expand the inner product for $i \in \mathbb{K} \setminus \{1, K\}$, yielding

$$
\begin{aligned}
&i \notin I : |\langle \psi_i, \Phi x_{c,p,\sigma} + r \rangle| \leq c(S+1)\alpha_{\min} - \left| \sum_{j \neq k} \sigma(j) c(p(j)) \langle \psi_i, \phi_j \rangle \right|, \\
&i \in I : |\langle \psi_i, \Phi x_{c,p,\sigma} + r \rangle| \geq c(S)\alpha_{\min} - \left| \sum_{j \neq i} \sigma(j) c(p(j)) \langle \psi_i, \phi_j \rangle \right|, \\
&\mathcal{I} \in \mathcal{A}_{1K+} : \\
&\qquad |\langle \psi_1, \Phi x_{c,p,\sigma} + r \rangle| \geq c(S)\alpha_{\min} n_{1+K} - \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right|, \\
&\mathcal{I} \in \mathcal{A}_{1\vee K+} : \\
&\qquad |\langle \psi_1, \Phi x_{c,p,\sigma} + r \rangle| \geq c(S)(\alpha_{\min} \frac{n_{1+K}}{2} + \beta_1 \frac{n_{1-K}}{2}) - \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right|, \\
&\mathcal{I} \in \mathcal{A}_{1\vee K-} : \\
&\qquad |\langle \psi_1, \Phi x_{c,p,\sigma} + r \rangle| \geq [c(S) - c(S+1)]\alpha_1 \frac{n_{1+K}}{2} \\
&\qquad\qquad\qquad + [c(S) + c(S+1)]\beta_1 \frac{n_{1-K}}{2} - \left| \sum_{j \notin \{1,K\}} \sigma(j) c(p(j)) \langle \psi_1, \phi_j \rangle \right|,
\end{aligned}
$$

Thus we achieve recovery of I, if for all $i \in \mathbb{K}$

$$
\max_{i \in \mathbb{K}} \left| \sum_{j \neq i} \sigma(j) c(p(j)) \langle \psi_i, \phi_j \rangle - c_{1K} \right| < \theta \cdot c(S)\alpha_{\min},
$$

where θ ensures again weaker conditions as in (22), namely

$$
\begin{aligned}
&\alpha_1 : \quad \frac{n_{1+K}}{2}(c(S) - c(S+1))\alpha_1 + \frac{n_{1-K}}{2}(c(S) + c(S+1))\beta_1 - \theta c(S)\alpha_{\max} \overset{!}{\geq} c(S+1)\alpha_{\max} + \theta c(S)\alpha_1, \\
&\alpha_{\min} : \quad c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \overset{!}{\geq} c(S+1)\alpha_{\max}, +\theta c(S)\alpha_{\min}.
\end{aligned}
$$

We continue by estimating the probability (31). We follow the same approach as in step 1, yielding

$$
\begin{aligned}
&\mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I_{1K} \vee \operatorname{sign}\left(\langle \psi_1, y \rangle\right) \neq \sigma(1) = \sigma(K)\right) \wedge 1, K \in I\right\}\right) \\
&\quad \leq \sum_{i \in \mathbb{K} \setminus \{K\}} \mathbb{P}\left(\left\{(I,\sigma,c) : \left| \sum_{j \neq i} \sigma(j) c(p(j)) \langle \psi_i, \phi_j \rangle - c_{1K} \right| \geq \theta c(S)\alpha_{\min} \right\} \cap \mathcal{A}_{1K+} \right). \qquad (33)
\end{aligned}
$$

We rewrite the probabilities as conditional probabilities, use the fact that every permutation is equally likely and split the sum into two parts; one over $j \in I \backslash \{i, 1, K\}$, that captures most of the energy, and the other over $j \in (I^c \cup \{1,K\}) \setminus \{i\}$. For $m_1 \in (0,1)$ and $m_2 = 1 - m_1$ we have

$$
\begin{aligned}
&\mathbb{P}\left(\left|\sum_{j\neq i} \sigma(j)c(p(j))\left\langle \psi_i, \phi_j\right\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min}, \mathcal{A}_{1K+}\right) \\
&\quad\leq \sum_{(\ell_1,\ell_K)\in\mathbb{S}^2} \mathbb{P}\left(\left|\sum_{j\neq i} \sigma(j)c(p(j))\left\langle \psi_i, \phi_j\right\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \mid (p(1),p(K)) = (\ell_1,\ell_K), \sigma(1)=\sigma(K)\right) \times \\
&\quad\quad \times \mathbb{P}((p(1),p(K)) = (\ell_1,\ell_K), \sigma(1)=\sigma(K)) \qquad (34) \\
&\quad\leq \frac{1}{2K(K-1)} \times \\
&\quad\quad \times \sum_{(\ell_1,\ell_K)\in\mathbb{S}^2} \mathbb{P}\left(\left|\sum_{j\in I\setminus\{i,1,K\}} \sigma(j)c(p(j))\left\langle \psi_i, \phi_j\right\rangle - c_{1K}\right| \geq m_1\theta c(S)\alpha_{\min}\right. \\
&\qquad\qquad\qquad\qquad \left.\mid (p(1),p(K)) = (\ell_1,\ell_K), \sigma(1)=\sigma(K)\right) \\
&\qquad\qquad + \mathbb{P}\left(\left|\sum_{j\in(I^c\cup\{1,K\})\setminus\{i\}} \sigma(j)c(p(j))\left\langle \psi_i, \phi_j\right\rangle - c_{1K}\right| \geq m_2\theta c(S)\alpha_{\min}\right. \\
&\qquad\qquad\qquad\qquad \left.\mid (p(1),p(K)) = (\ell_1,\ell_K), \sigma(1)=\sigma(K)\right).
\end{aligned}
$$

For the first term we use Lemma 4 while for the second term we use Hoeffding's inequality. With some small simplifications we get for all $i \in \mathbb{K} \setminus \{K\}$,

$$
\begin{aligned}
&\mathbb{P}\left(\left|\sum_{j\neq i} \sigma(j)c(p(j))\left\langle \psi_i, \phi_j\right\rangle - c_{1K}\right| \geq \theta c(S)\alpha_{\min} \mid (p(1),p(K)) = (\ell_1,\ell_K), \sigma(1)=\sigma(K)\right) \\
&\quad\leq 2\exp\left(\frac{-\left(m_1\theta c(S)\alpha_{\min}\right)^2}{2\left(c(1)\hat{\mu}\cdot m_1\theta c(S)\alpha_{\min} + \|c(\mathbb{S})\|_2^2\frac{B}{K-S}\right)}\right) + 2\exp\left(\frac{-\left(m_2\theta c(S)\alpha_{\min}\right)^2}{2\hat{\mu}^2\left(c(\ell_1)^2 + c(\ell_K)^2 + \|c\left(\mathbb{S}^c\right)\|_2^2\right)}\right) \\
&\quad\leq 2\exp\left(-\frac{1}{4}\min\left\{\frac{c(S)m_1\theta\alpha_{\min}}{c(1)\hat{\mu}}, \frac{(K-S)\left(m_1\theta c(S)\alpha_{\min}\right)^2}{B\|c(\mathbb{S})\|_2^2}\right\}\right) \\
&\quad\quad + 2\exp\left(-\frac{1}{4}\min\left\{\frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{2c(1)^2\hat{\mu}^2}, \frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{\hat{\mu}^2\|c\left(\mathbb{S}^c\right)\|_2^2}\right\}\right)
\end{aligned}
$$

Hence we get the following upper bound:

$$
\begin{aligned}
&\mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I_{1K} \vee \operatorname{sign}\left(\langle\psi_1, y\rangle\right) \neq \sigma(1) = \sigma(K)\right) \wedge 1, K \in I\right\}\right) \\
&\quad\leq 2\frac{(K-1)S(S-1)}{2K(K-1)}\exp\left(-\frac{1}{4}\min\left\{\frac{c(S)m_1\theta\alpha_{\min}}{c(1)\hat{\mu}}, \frac{(K-S)\left(m_1\theta c(S)\alpha_{\min}\right)^2}{B\|c(\mathbb{S})\|_2^2}\right\}\right) \\
&\quad\quad + 2\frac{(K-1)S(S-1)}{2K(K-1)}\exp\left(-\frac{1}{4}\min\left\{\frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{2c(1)^2\hat{\mu}^2}, \frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{\hat{\mu}^2\|c\left(\mathbb{S}^c\right)\|_2^2}\right\}\right),
\end{aligned}
$$

where θ has to ensure (23) and $m_1 + m_2 = 1$. So whenever $\alpha_{\min}$ and α_1 fulfill (26) we get that

$$
\mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I_{1K} \vee \operatorname{sign}\left(\langle\psi_1, y\rangle\right) \neq \sigma(1) = \sigma(K)\right) \wedge 1, K \in I\right\}\right) \leq 2\frac{(K-1)S(S-1)}{K(K-1)} \cdot K^{-n}. \qquad (35)
$$

In the same manner we proceed with the other cases of (32), except for the most problematic case below and for $\mathbb{P}\left(\left\{(I,\sigma,c) : 1 \in \hat{I} \wedge 1, K \notin I\right\}\right)$:

$$\begin{aligned}&\mathbb{P}\left(\left\{(I,\sigma,c) : \left(\hat{I} \neq I_{1K} \vee \operatorname{sign}\left(\langle \psi_1, y\rangle\right) \neq \sigma(1) \neq \sigma(K)\right) \wedge 1, K \in I\right\}\right) \\ &\quad \leq \binom{K-2}{S-2} \Big/ \left(2 \cdot \binom{K}{S}\right) = \frac{S(S-1)}{2K(K-1)}. \end{aligned} \tag{36}$$

For $\mathbb{P}\left(\left\{(I,\sigma,c) : 1 \in \hat{I} \wedge 1, K \notin I\right\}\right)$ we easily verify that the following weaker condition compared to (23) needs to be fulfilled

$$c(S)\alpha_{\min} - \theta c(S)\alpha_{\min} \overset{!}{\geq} c(S+1)\left(\alpha_1 n_{1+K} + \beta_1 n_{1-K}\right) + \theta c(S)\alpha_{\min}.$$

So we can estimate similar as in step 2, yielding

$$\begin{aligned}&\mathbb{P}\left(\left\{(I,\sigma,c) : 1 \in \hat{I} \wedge 1, K \notin I\right\}\right) \\ &\quad \leq \sum_{(\ell_1,\ell_K)\in(\mathbb{K}\setminus\mathbb{S})^2} \mathbb{P}\left(1 \in \hat{I} \mid (p(1), p(K)) = (\ell_1, \ell_K)\right) \cdot \mathbb{P}\left((p(1), p(K)) = (\ell_1, \ell_K)\right) \\ &\quad \leq \frac{1}{K(K-1)} \times \\ &\qquad \times \sum_{(\ell_1,\ell_K)\in(\mathbb{K}\setminus\mathbb{S})^2} \sum_{i \in I \cup \{1\}} \mathbb{P}\left(\left|\sum_{j \neq i} \sigma(j) c(p(j)) \langle \psi_i, \phi_j\rangle\right| \geq \theta \cdot c(S)\alpha_{\min} \mid (p(1), p(K)) = (\ell_1, \ell_K)\right).\end{aligned}$$

Using the same splitting technique as in step 2, we get

$$\begin{aligned}&\mathbb{P}\left(\left\{(I,\sigma,c) : 1 \in \hat{I} \wedge 1, K \notin I\right\}\right) \\ &\quad \leq 2\frac{(S+1)(K-S)(K-S-1)}{K(K-1)} \exp\left(-\frac{1}{4}\min\left\{\frac{c(S)m_1\theta\alpha_{\min}}{c(1)\hat{\mu}}, \frac{(K-S)\left(m_1\theta c(S)\alpha_{\min}\right)^2}{B\|c(\mathbb{S})\|_2^2}\right\}\right) \\ &\qquad + 2\frac{(S+1)(K-S)(K-S-1)}{K(K-1)} \exp\left(-\frac{1}{4}\min\left\{\frac{\left(c(S)m_2\theta\alpha_{\min}\right)^2}{\hat{\mu}^2\left\|c\left(\mathbb{S}^c\right)\right\|_2^2}\right\}\right)\end{aligned}$$

From this, whenever

$$\begin{aligned}\alpha_{\min} \geq \max\bigg\{&\frac{1}{1-2\theta}\frac{c(S+1)}{c(S)}(\alpha_1 n_{1+K} + \beta_1 n_{1-K}), \frac{4n}{m_1\theta_1}\frac{c(1)}{c(S)}\hat{\mu}\log K, \\ &\frac{2\sqrt{n}}{m_1\theta_1}\frac{\|c(\mathbb{S})\|_2}{c(S)}\sqrt{\frac{B\log K}{K-S}}, \frac{2\sqrt{n}}{(1-m_1)\theta}\frac{\left\|c\left(\mathbb{S}^c\right)\right\|_2}{c(S)}\hat{\mu}\sqrt{\log K}\bigg\}\end{aligned}$$

we get that

$$\mathbb{P}\left(\left\{(I,\sigma,c) : 1 \in \hat{I} \wedge 1, K \notin I\right\}\right) \leq 4\frac{(S+1)(K-S)(K-S-1)}{K(K-1)} \cdot K^{-n}.$$

Combining the bound above with the ones in (31) and (36) as well as using that $\frac{S-\ell}{K-\ell} \leq \frac{S}{K}$ for $\ell < S$, we obtain

$$\mathbb{P}(\mathcal{R}_{1K}) \leq \left[\frac{2S(S-1)}{K} + \frac{8S(K-S)}{K} + \frac{4(S+1)(K-S)^2}{K}\right] \cdot K^{-n} + \frac{S^2}{2K^2}.$$

Putting it all together:
We start by summarizing all assumptions on $\alpha_{\min}$ and α_1, yielding

$$\alpha_{\min} \geq \max\left\{\frac{1}{1-2\theta}\frac{c(S+1)}{c(S)}\max\{\alpha_{\max}, \alpha_1 n_{1+K} + \beta_1 n_{1-K}\}, \frac{4n}{m_1\theta_1}\frac{c(1)}{c(S)}\hat{\mu}\log K,\right.$$
$$\left.\frac{2\sqrt{n}}{m_1\theta}\frac{\|c(\mathbb{S})\|_2}{c(S)}\sqrt{\frac{B\log K}{K-S}}, \frac{6\sqrt{n}}{(1-m_1)\,\theta}\frac{c(1)}{c(S)}\hat{\mu}\sqrt{\log K}, \frac{2\sqrt{n}}{(1-m_1)\,\theta}\frac{\|c\,(\mathbb{S}^c)\|_2}{c(S)}\hat{\mu}\sqrt{\log K}\right\} \quad \text{and}$$
$$\alpha_1 \geq \frac{2}{n_{1+K}\cdot(1-\frac{c(S+1)}{c(S)})-4\theta}\frac{c(S+1)}{c(S)}\alpha_{\max} - \frac{n_{1-K}}{2}c(S)\beta_1.$$

Setting $\theta = \frac{5}{16}, m_1 = \frac{2}{3}$ and $n = 3$, the probability that thresholding fails to recover I or the corresponding signs, restricted to the signals for which we have $k \in I \setminus \{1, K\}$, is bounded by

$$\mathbb{P}\left(\left\{(I,\sigma,c) : \left[\left(\hat{I} \neq I \vee \operatorname{sign}\left(\langle\psi_k, y\rangle\right) \neq \sigma(k)\right) \wedge k \in I\right] \vee \left(k \in \hat{I} \wedge k \notin I\right)\right\}\right)$$
$$\leq \left[S\left(\frac{2S^2}{K^2} + \frac{8S(K-S)}{K^2} + \frac{4(K-S)^2}{K^2}\right)\right.$$
$$\left. +(S+1)\left(\frac{2S^2(K-S)}{K^3} + \frac{8S(K-S)^2}{K^3} + \frac{4(K-S)^3}{K^3}\right)\right]\cdot K^{-3} + \frac{S^3}{2K^3}$$
$$\leq 8(S+1)K^{-3} + \frac{S^3}{2K^3} = \frac{16(S+1)+S^3}{2K^3} \quad \text{and}$$
$$\mathbb{P}(\mathcal{R}_{1K}) \leq \left[\frac{2S(S-1)}{K} + \frac{8S(K-S)}{K} + \frac{4(S+1)(K-S)^2}{K}\right]\cdot K^{-3} + \frac{S^2}{2K^2}.$$
$$\leq \frac{S^2}{K^2}\left(\frac{2}{K^2} + \frac{12}{KS} + \frac{1}{2}\right).$$

whenever

$$\alpha_{\min} \geq \max\left\{\frac{8}{3}\frac{c(S+1)}{c(S)}\max\{\alpha_{\max}, \alpha_1 n_{1+K} + \beta_1 n_{1-K}\}, 58\frac{c(1)}{c(S)}\hat{\mu}\log K, \frac{50}{3}\frac{c(1)}{c(S)}\sqrt{\frac{SB\log K}{K-S}}\right\} \text{and}$$
$$\alpha_1 \geq \frac{2}{n_{1+K}\cdot(1-\gamma_{gap})-\frac{20}{16}}\cdot\gamma_{gap}\cdot\alpha_{\max} - \frac{n_{1-K}}{2}c(S)\beta_1.$$

where we have used that $\|c(\mathbb{S})\|_2 \leq \sqrt{S}c(1)$, $\log K \geq 7$ and $\frac{\|c(\mathbb{S}^c)\|_2}{c(1)} \leq \frac{12}{7}\sqrt{\log K}$.

■

Lemma 4 (Proposition 5 in [19]) *Let $v \in \mathbb{R}^K$ be a vector, $I = (i_1, \ldots, i_S)$ be a sequence of length S obtained by sampling from $\mathbb{K} = \{1, \ldots, K\}$ without replacement, ε with values in $\{-1,1\}^S$ a Rademacher vector independent from I and $c \in \mathbb{R}^S$ a scaling vector. Then for any $t \geq 0$,*

$$\mathbb{P}\left(\left|\sum_{k=1}^{S} c_k\varepsilon_k v_{i_k}\right| \geq t\right) \leq 2\exp\left(\frac{-t^2}{2\left(\|c\|_\infty\|v\|_\infty t + \|c\|_2^2\cdot\|v\|_2^2/(K-S)\right)}\right).$$

Lemma 5 (B.6. in [24]) *For $y_n = \Phi x_{c_n,,p_n,,\sigma_n}$ as in the model of 2 and $B := \|\Phi\|_{2,2}^2$ we have*

$$\mathbb{P}\left(\left|\frac{1}{N}\sum_n \mathbb{1}_{I_n}(i)\sigma_n(i)\left\langle y_n, \phi_i\right\rangle\right| \leq (1-t)\frac{\gamma_{1,S}}{K}\right) \leq \exp\left(-\frac{Nt^2\gamma_{1,S}^2}{2K\left(1+\frac{SB}{K}+t\gamma_{1,S}\sqrt{B}/3\right)}\right), \tag{37}$$

Proof As in [24] Lemma B.6., but since in our model we have no noise we can estimate $|\langle y, \phi_k \rangle| \leq \|y\|_2 \leq \sqrt{B}$, instead of $\sqrt{B+1}$. ■

Lemma 6 (B.10. in [24]) *If for two vectors ψ, ϕ, where $\|\phi\|_2 = 1$, and two scalars $0 < t < s$ we have, $\|\psi - s\phi\|_2^2 \leq t^2$ then*

$$\left\| \frac{\psi}{\|\psi\|_2} - \phi \right\|_2^2 \leq 2 - 2\sqrt{1 - \frac{t^2}{s^2}}.$$

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